SOFTWARE DATA ANALYSIS TOOLS FOR TLS AND LAMS

ADVANCED NAVIGATION & POSITIONING CORPORATION

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Supplement for ANPC analysis tools

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Software data analysis tools for TLS and LAMS

Overview

The use of three software tools will be explained below, examples of data are provided and some practical troubleshooting is described. The example data for each tool includes first a sample of normal data followed by data that shows a common troubleshooting that can be facilitated using the tool.

Software tools explained within this document:

1. RAM
2. MultiBIT
3. Look

Using the RAM software tool

The RAM tool [Figure 4-33] is used to graphically display data contained in a TLS .ram file. RAM files are generated by the TLS base software and contain raw sensor data recorded during BIT or in Test mode and Track Mode. The tool consists of a plot window and several pull-down menus and toolbars. The pull-down menus and toolbars are used to control the display of sensor data in the plot window. In addition, any part of the plots may be zoomed in or out vertically and/or horizontally through mouse and keyboard controls. The primary function of the RAM tool is to graphically display the digitized sensor data, providing the user with a means to perform troubleshooting tasks. The tool provides the ability to inspect the transponder replies in track ram files.

Bit ram files are collected when the maintenance interface setting dialogue check box for bit ram is selected. BIT ram files are used to verify the position of the cal/bit pulse within the AOA sensor data window and verify the quality of the cal/bit pulse.

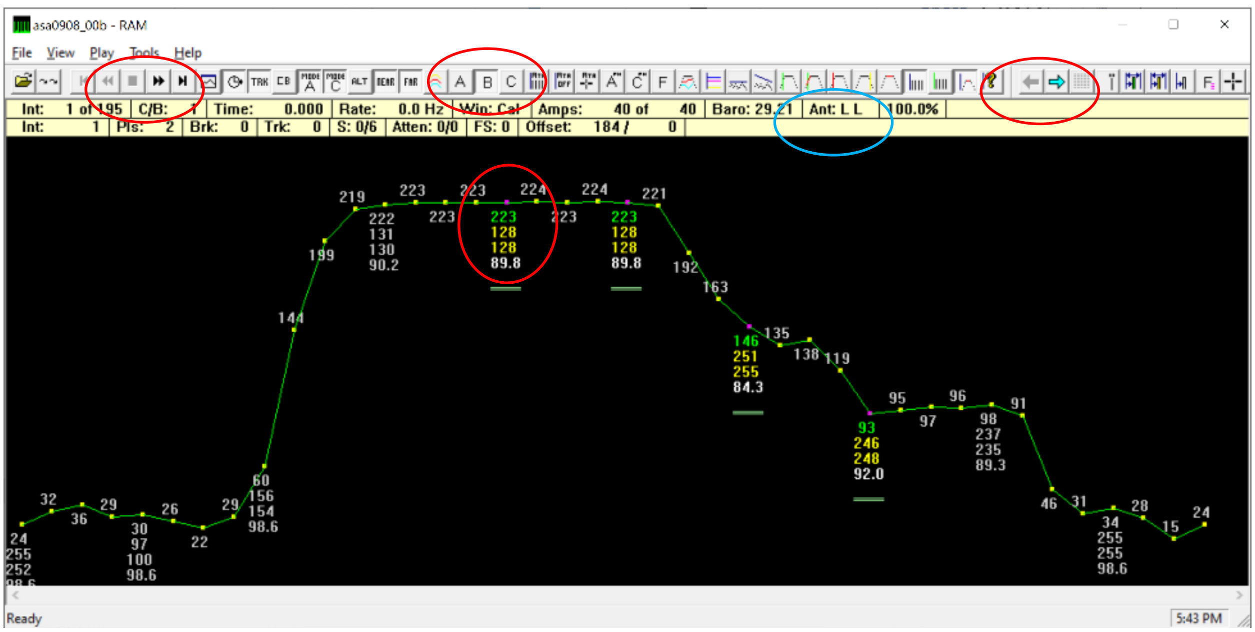

Figure 1 - Representative of a normal Cal/BIT pulse for a properly installed TLS

The pulse should be square, without evidence of a step in peak amplitude that indicates that there is no short-path multipath arriving at the antenna that would cause degradation of the signal. Short-path multipath is the most destructive type of signal interference because it is coherent (same frequency) and combines with the direct path to distort the signal. The figure above shows shape characteristics of an ideal cal/bit pulse. In the example above all of the pulse peak amplitudes are within 2 counts. Referring to the numbers at the center of the pulse, 223/224 is the amplitude, 89.8 is the frequency with the leading 10 omitted (so 1089.8 was the measured frequency).

For systems that use the 920-00256 aoa sensor, the A and C buttons allow displaying the individual channels. The phase measurement of 128 counts was the measured phase difference on the Low channel (channel indicated by “Ant: LL” circled blue). Note that ”Ant” is the abbreviation for “antenna” and the antenna that is connected to Channel A or C corresponds to the first letter of the antenna name, L for Low, M for Medium and H for High.

Like all TLS SW tools, hovering the cursor over the button will reveal the name of the button. The samples can be examined by pressing the fast forward button to replay the entire file or by pressing the blue arrows to go forward or backward through the file one update at a time.

In the figure below, shortpath multipath arrived at the antenna and combined in phase with the signal that arrived directly from the cal/bit, causing corruption of the pulse shape that can be observed as a step in the peak amplitude of the pulse (circled red).

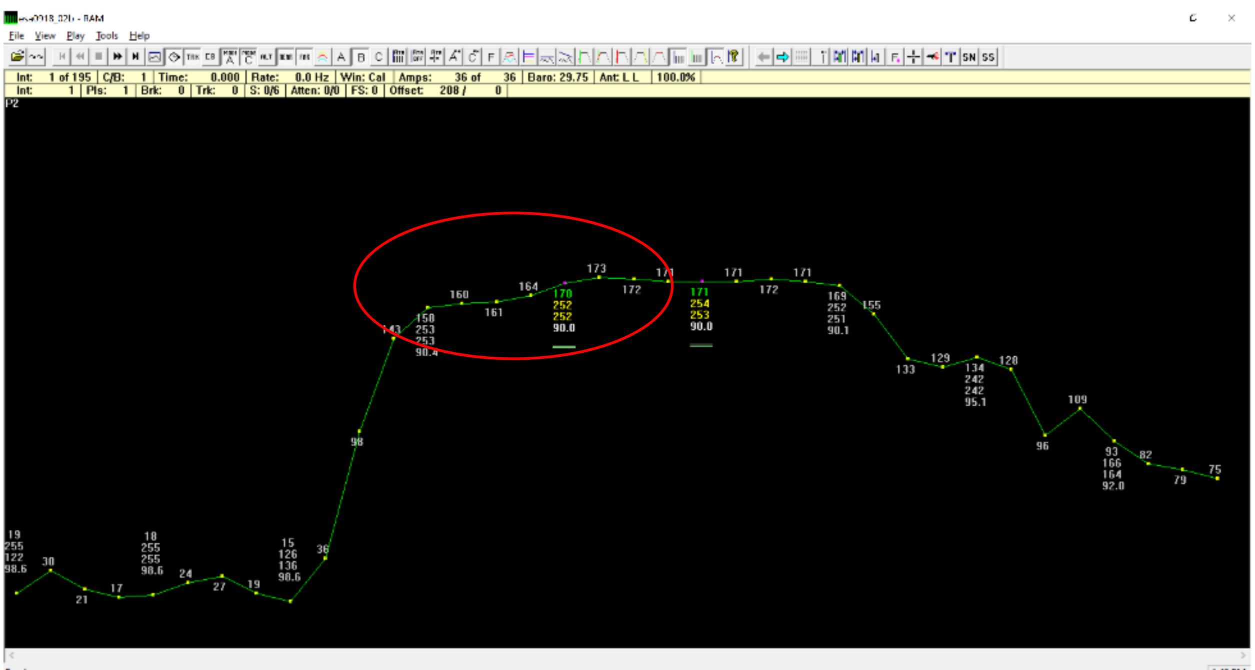

Figure 2 - Short path multipath interference

In the later part of the bit cycle that windows out, the scale of the x axis makes the corrupted pulse shape more observable. The source of the reflection in this case must be identified and eliminated or the cal/BIT and sensor moved to eliminate this symptom. If this multipath condition was to remain, the measured TOA and phase will not remain stable and will cause false alarms that reduce system availability. The term "false alarm" is used here to describe a parameter that exceeds tolerance because of a configuration problem rather than an actual hardware problem that would impact on the accuracy of the transmitted localizer or glide slope.

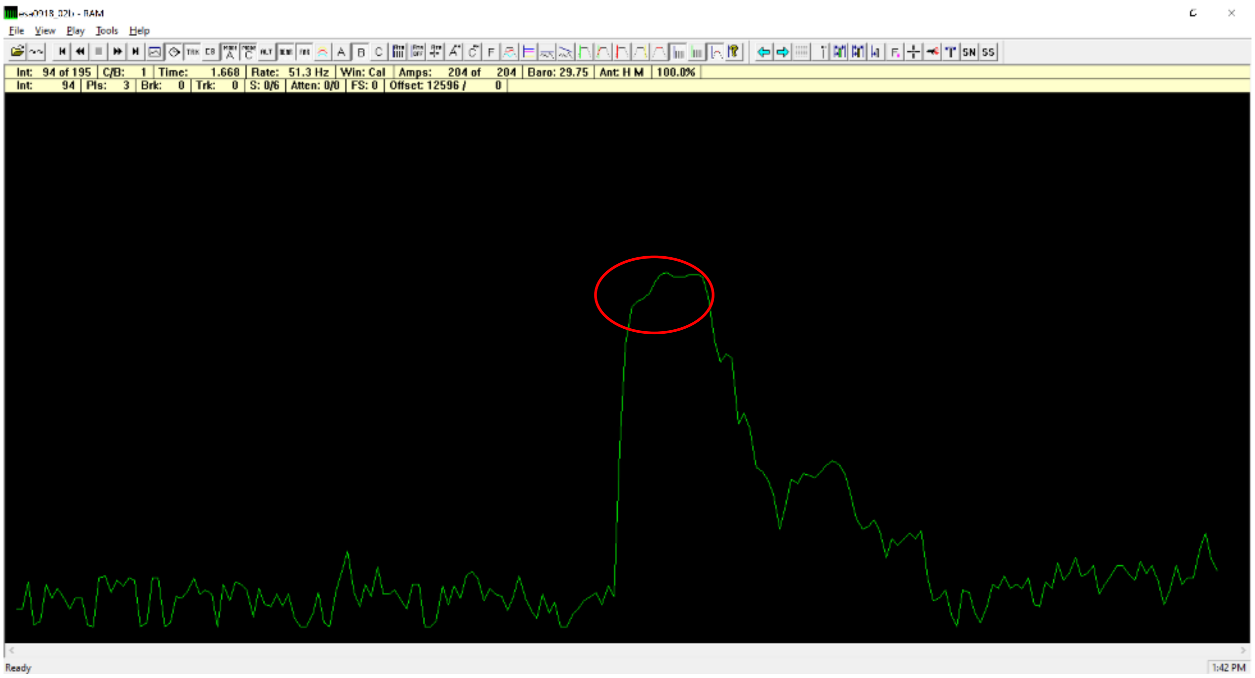

Figure 3 - short path multipath interference

The ram tool allows examination of individual sensor data channels to verify that each channel is operating normally. In the figure below, the ATA channel has been selected for display on the ASA sensor and the ATA pulse is normally offset in time from the other channels since there is a large physical separation of that antenna from the ASA antennas.

The Ram tool is used to examine the Cal/BIT pulse amplitudes in order to assess if an attenuator must be added to the Cal/BIT and/or ATA signal path in order to properly set the amplitude monitor limits, TI manual 020-00073 includes tunning of the cal/bit signal amplitude.

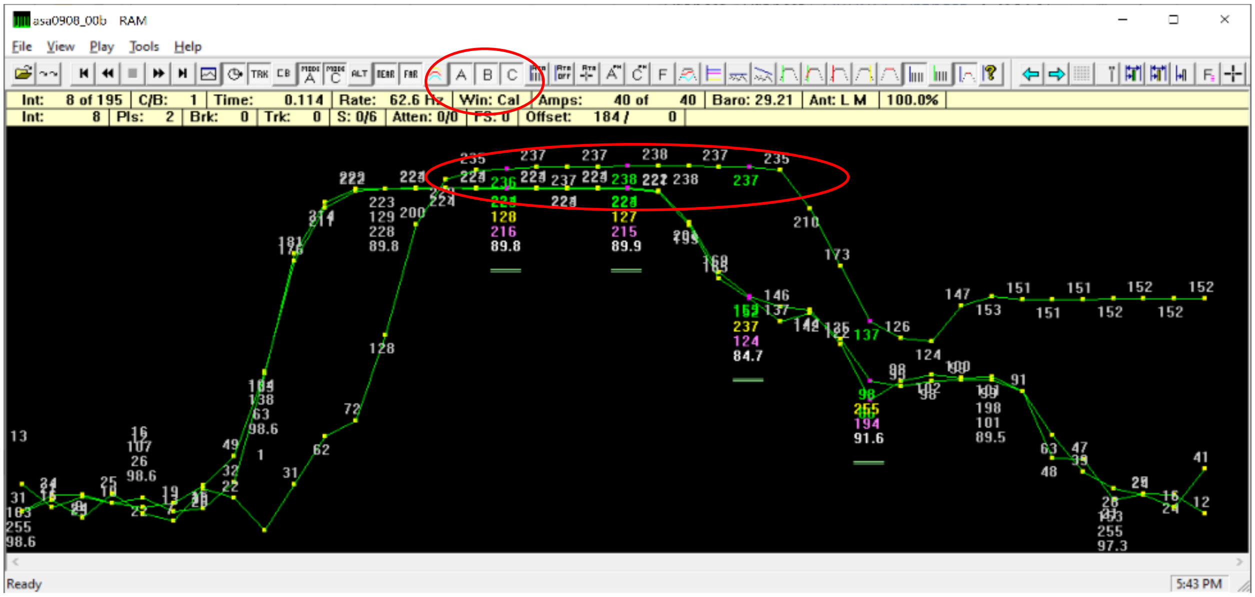

Figure 4 - verifying pulse amplitude during install

The figure below shows how a long-path multipath reflection appears in the Ram buffer, these do not interfere with the stability of the cal/bit pulse and can be ignored.

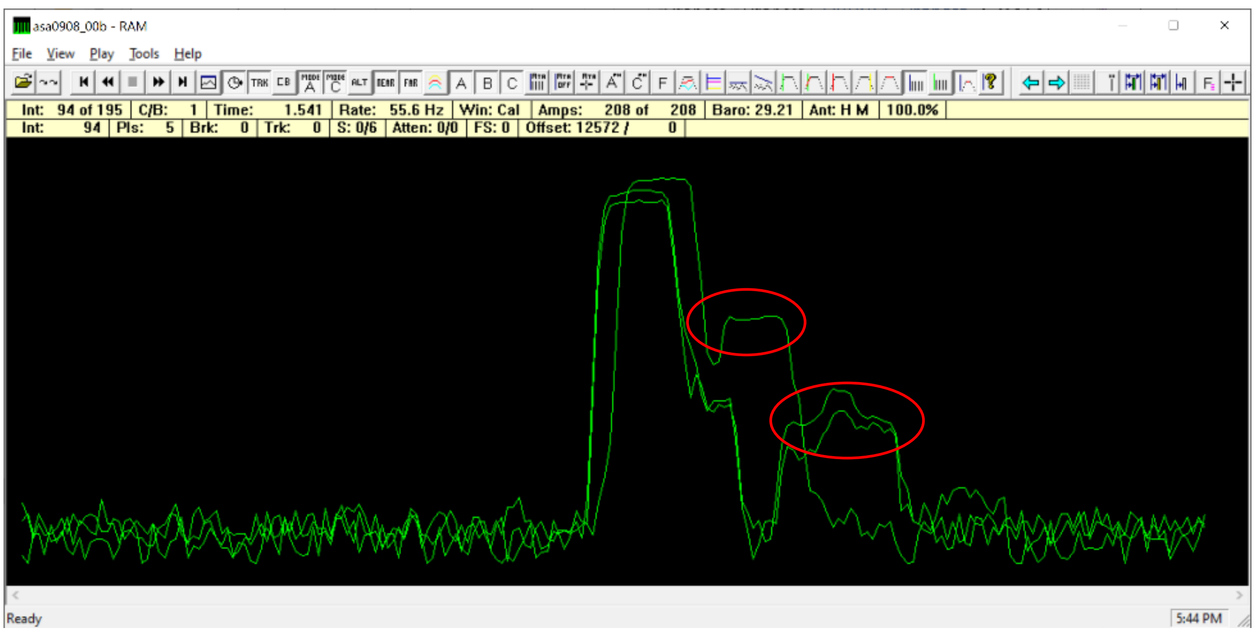

Figure 5 - long path multipath

Track ram

Track ram files are collected when the setting dialogue check box for track ram is selected. Track ram can be used to verify that synchronous transponder replies are returning to the system aoa sensors. Track ram can also be used to any particular transponder reply parameters.

As mentioned above for the bit/ram, it is possible to zoom in on individual aircraft transponder replies to inspect various pulse characteristics for spec compliance. The bracket pulse spacing or pulse width and frequency can all be verified. There have been occasions over the past 15 years when an aircraft transponder was replying out of tolerance. Examples include the frequency being offset from 1090 mhz by more than the tolerance, the pulse width was out of tolerance and in one case the F1 pulse was 10 microseconds in width when the nominal value is 450ns.

The ram file contains data for all aircraft within operable range of the system. The raw file (described in a section below) contains more detailed data for any one aircraft when the "acquire" button is pressed on the RCU.

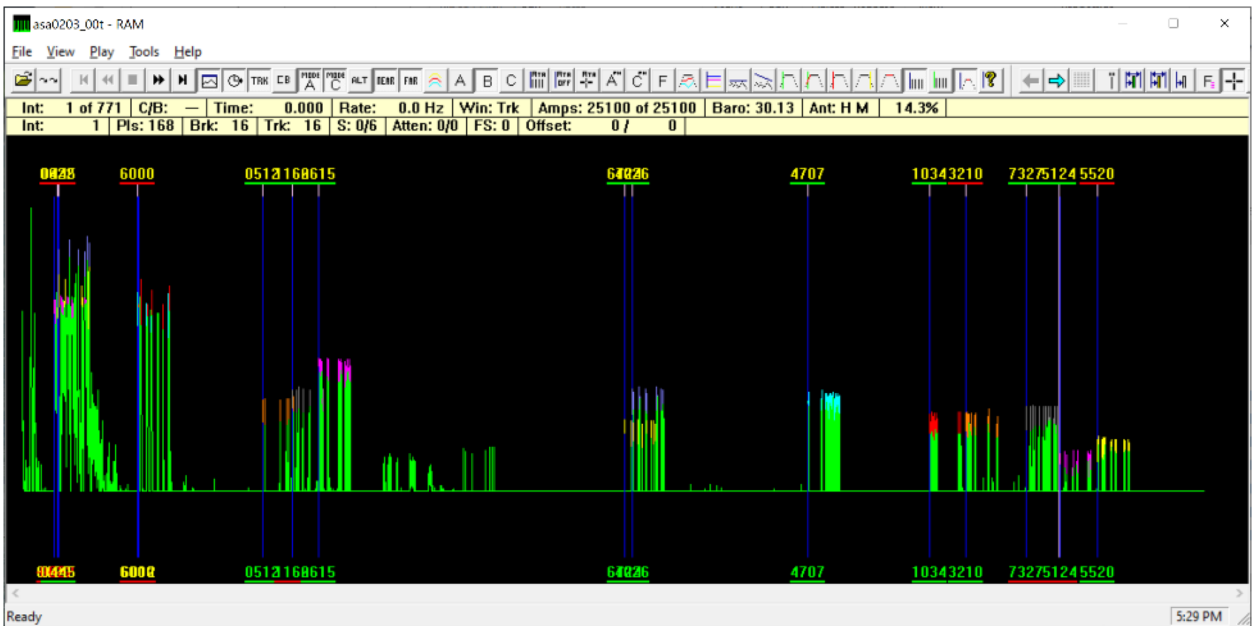

Figure 6 - track ram file

Using track ram, it is possible to see the cal/bit pulse and the two Mode 3/A interrogation pulses. The interrogation pulses are clipped since the aoa sensor filter is tuned for the transponder frequency of 1090 mhz, not the interrogation frequency of 1030 mhz.

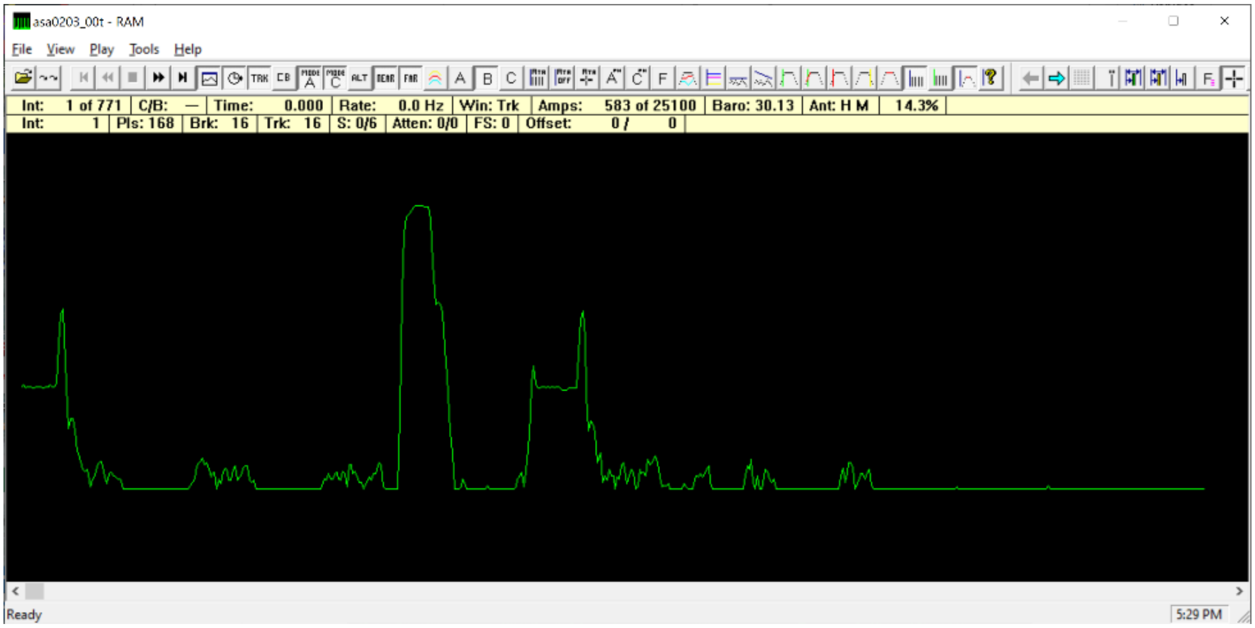

Figure 7 - interrogation pulses and near Cal/BIT pulse

The Ram tool has a rubber band tool for measuring pulse timing characteristics. Press ctrl left click to access rubber band tool, framing pulses are nominally space at 21.3 microseconds. Refer to the TLS system spec or ICAO Annex 10 for all other transponder timing tolerances if it is suspected that a transponder is out of tolerance.

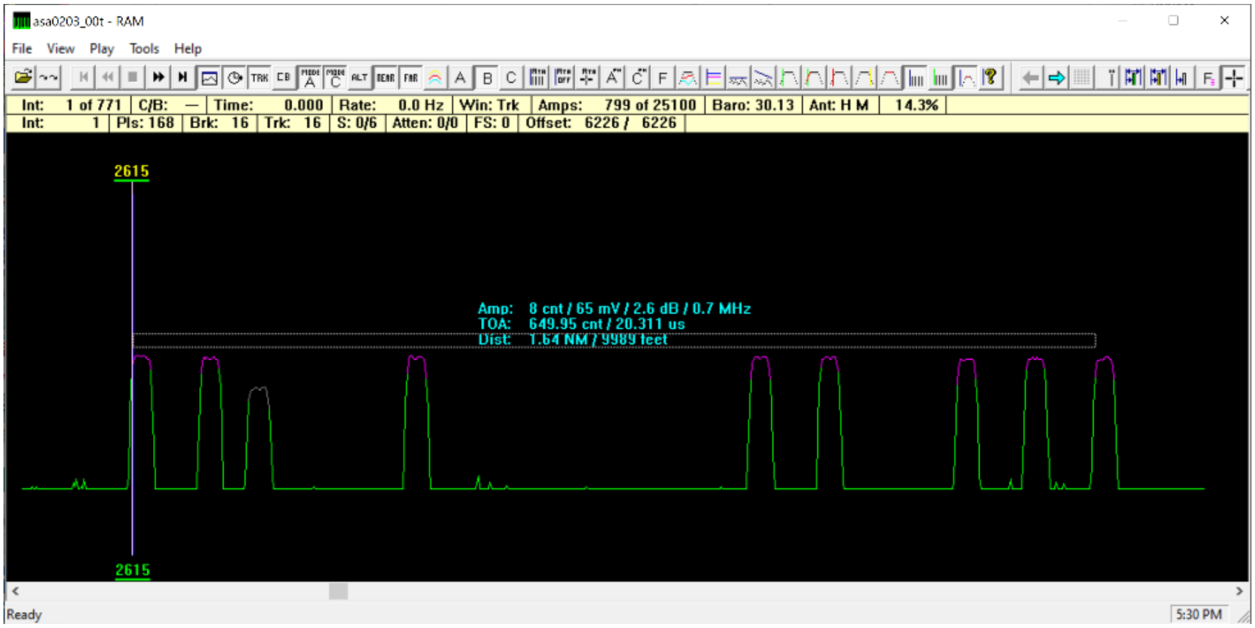

Figure 8 - measuring with the rubber band tool

Press ctrl left click to access rubber band tool, pulse width nominal are 450 ns +/- 100 ns for limits. Refer to the TLS system spec or ICAO annex 10 for all other timing tolerances if it is suspected that a transponder is out of tolerance.

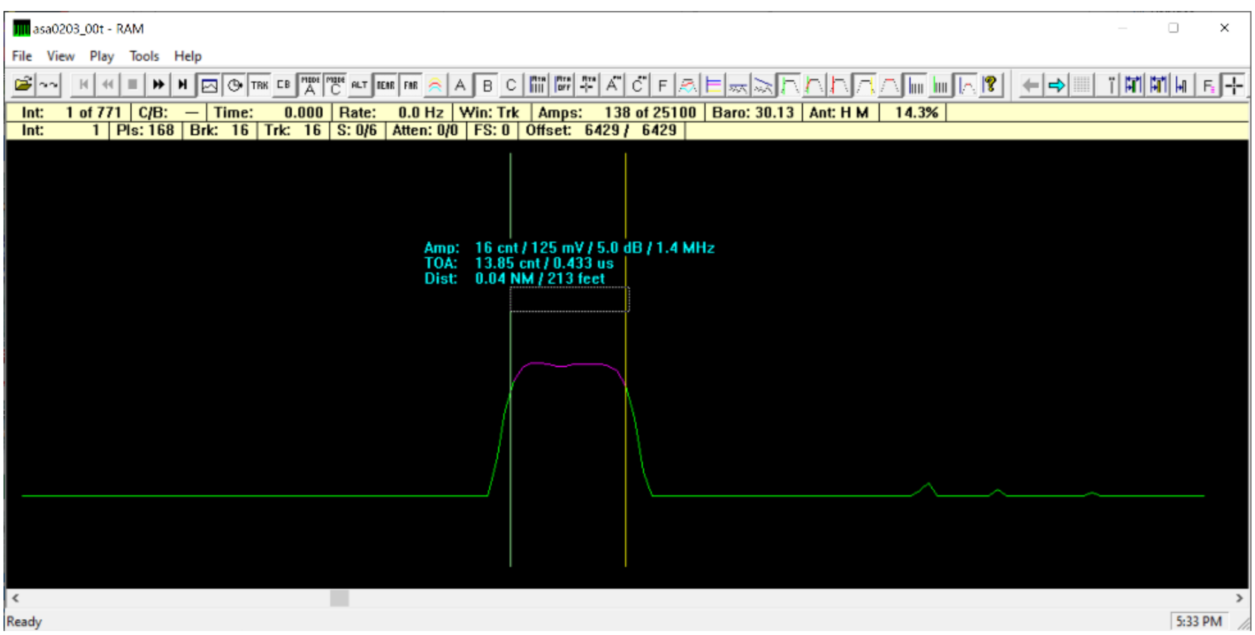

Figure 9 - Measuring pulse width with the rubber band tool

In the situation shown in the figure below, the transponder replies from the flight inspection aircraft squawking 0024 is suffering synchronous garble from replies by other area aircraft. In this case, the suppression pulse p2 was disabled in the maintenance interface Settings dialogue. Ensure that, P2 is enabled whenever possible to decrease the possibility that the aircraft conducting a TLS approach will suffer a reduced update rate caused by synchronous garble.

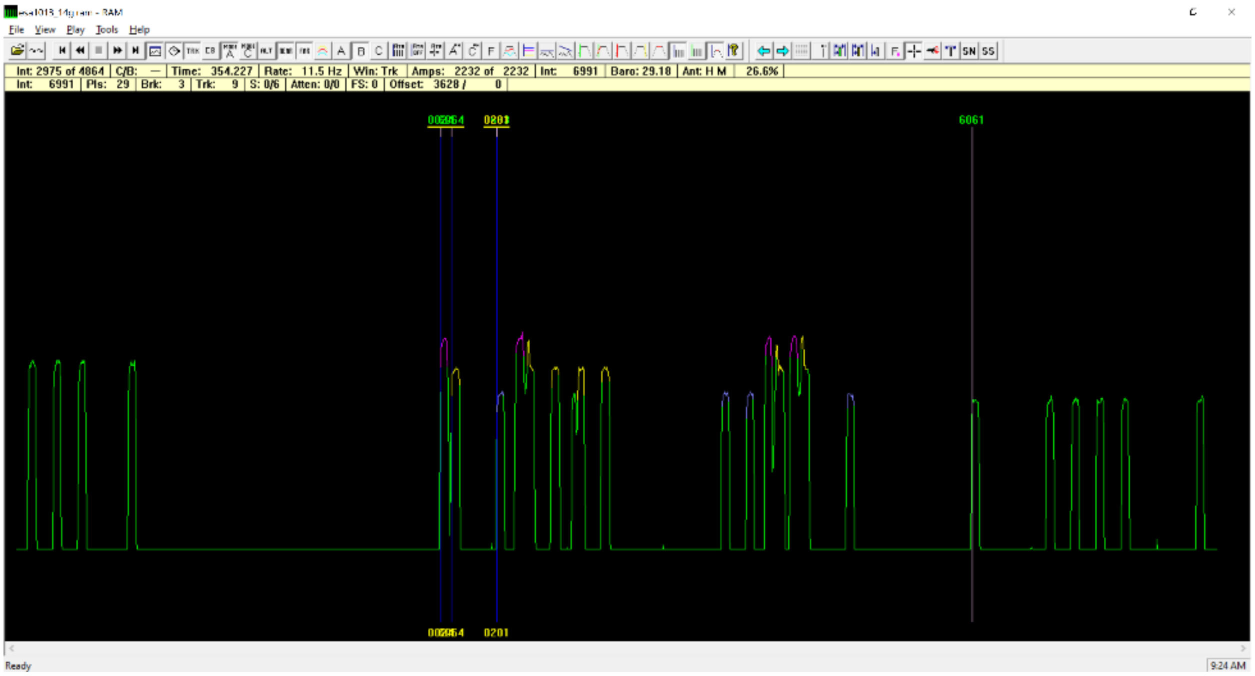

Figure 10 - Example of synchronous garble that reminds the user to enable the p2 side-lobe suppression pulse

Using the Multibit software tool

The MultiBIT tool is used to graphically display data contained in TLS BIT files (ESA.bit, ASA.bit, ATA.bit, uplink_*.bit and interrog.bit). These files are generated by the TLS base software and contain sensor health data collected during BIT. This data includes TOA, amplitude, phase, temperature and frequency measurements to name a few. The tool consists of a plot window and several pull-down menus and toolbars. The pull-down menus and toolbars are used to control the display of data in the plot window. In addition, any part of the plots may be zoomed in or out vertically and/or horizontally through mouse and keyboard controls. Be sure to make all system .dat files available in the parent folder, the monitor limits may be displayed when the system monitor_limits.dat file is in the directory where BIT or RAM files are selected or in its parent folder.

TLS BIT in best analyzed using the multibit tool. Please use the multibit tool to examine BIT data and do not try to use a text editor such as notepad to determine that the system is functioning properly and passing BIT. Like all ANPC SW tools, hovering the cursor over the toolbar button will reveal the name of the button and indicate the function of the button.

The system provides a BIT to monitor key performance parameters. The result of BIT is pass or fail. If a parameter is near 80% of the alarm limit then the parameter may cause and alert. Multibit is helpful to make the BIT cycle result less binary and understand any trends or problems that are developing prior to them resulting in a system outage.

The user should use multibit to examine all key performance parameters compared to the monitor limits. Referring to the figure below, the best way to review all parameters quickly and thoroughly is to select the switch symbol in the upper left section of the tool bar (circled red) and select the monitor limits button (circled green). With the switch symbol enabled, the look tool will toggle off the previous parameter when a new one is selected, this allows inspecting each parameter individually. With the switch and monitor limits selected, proceed from left to right across the tool bar displaying each parameter in turn and verify that all are well centered in the monitor limits or examine more closely any time periods when alerts limits or alarms limits were exceeded. Diagnose these events and determine corrective action.

Figure 11 - High resolution channel amplitude from the ASA showing normal data with no problems observed.

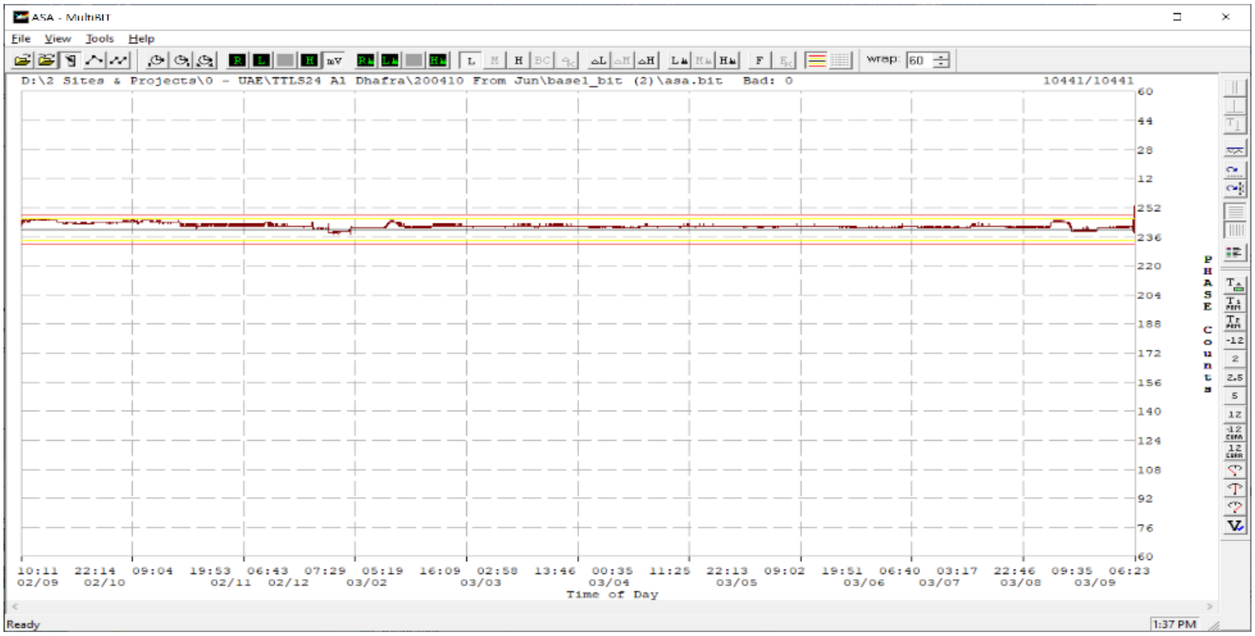

Figure 12 - Low resolution channel phase from the ASA showing a slight trend toward the upper alert limit.

The “Switch” button may be disabled to display multiple parameters simultaneously to check for correlation between one parameter and another. In the figure below, both the ASA high-resolution phase and the temperature are selected to view a temperature correlation in the phase variation.

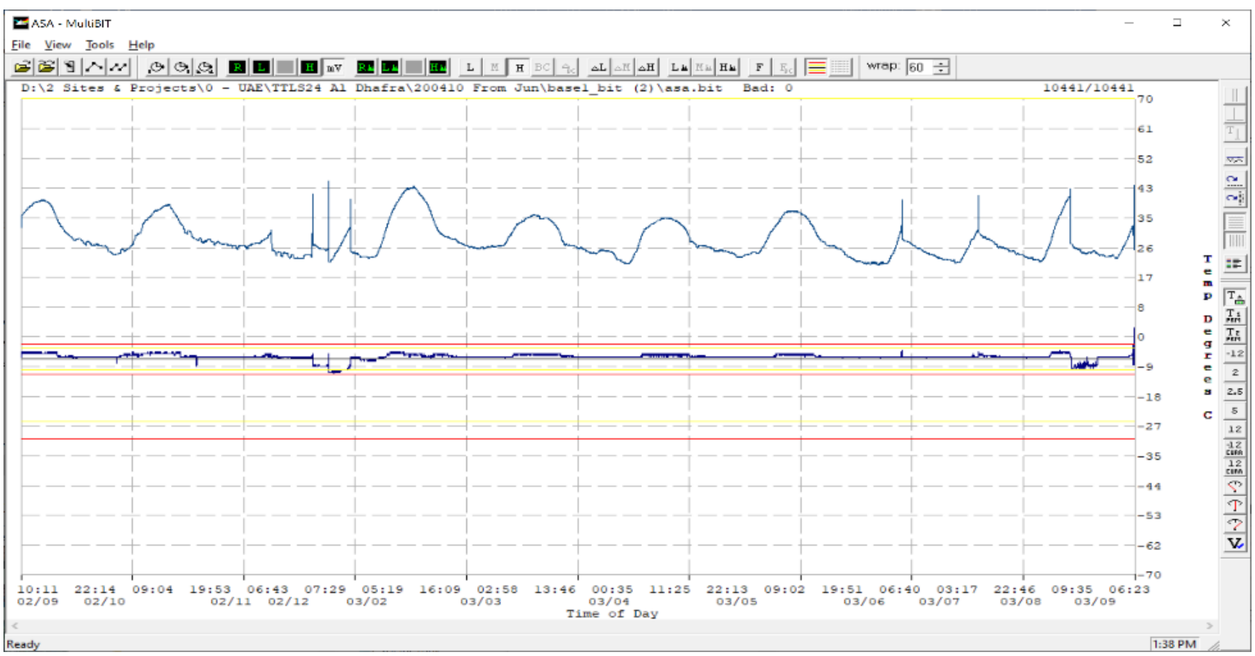

Figure 13 - High resolution channel phase measurement from the ASA showing normal variation as a function of diurnal temperature cycles (temperature in blue)

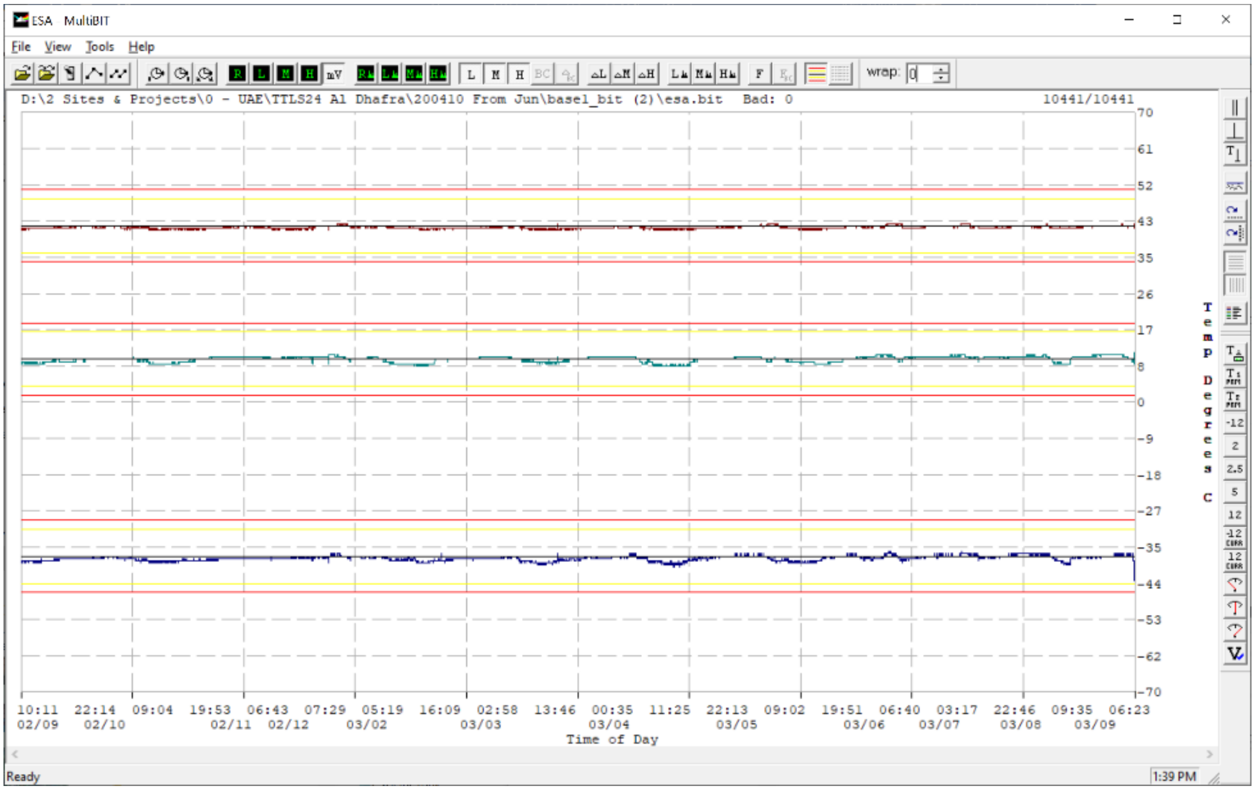

Figure 14 - High resolution channel phase measurement from the ESA showing normal daily variation.

Using the Look software tool

The Look tool below is used to graphically display data contained in a TLS .raw file. Raw files are generated by the TLS base software and contain processed sensor data collected for a guidance track. This data is the output of the TLS Pulse Processing software and the input to the TLS Track Processing software, and includes transponder and Cal/BIT data.

The tool consists of a plot window and several pull-down menus and toolbars. The pull-down menus and toolbars are used to control the display of raw file data in the plot window. In addition, any part of the plots may be zoomed in or out vertically and/or horizontally through mouse and keyboard controls. Data can be displayed for the ESA, ASA or both. The monitor limits may be displayed if the system monitor_limits.dat file is in the directory of the raw file selected or in its parent directory.

The TLS records measured data during an approach in the ram file mentioned above and then also the raw file described here. Each raw corresponds to pressing acquire for and aircraft, the data starts recording to the file when the aircraft is detected within the service volume. The following notes describe specific things to look for in the data file to observe normal operation and to be able to diagnose problems. Like all of the SW tools, hovering the cursor over the button will reveal the name of the button.

TOA

Shown in the figure below is the normal time-of-arrival measurement that results for an aircraft on TLS approach. The range from runway threshold is displayed on the left vertical axis and horizontal lower axis. The units can be adjusted with the axis buttons circled in red. The missing TOA measurements are displayed as vertical lines. They indicate where the transponder did not reply to TLS because the transponder was occupied by replying to another radar system. Normal rates of missed measurements range for 5 to 15%. These missing measurements can also be caused by synchronous garble where multiple transponder replies arrive simultaneously at the TLS sensor antenna.

In the example below, since the TOA measurement peaks at 12.2 nm, we know the aircraft turned inbound toward the TLS at 12.2 nm and the file ends when the aircraft nears the TLS and crosses the runway threshold.

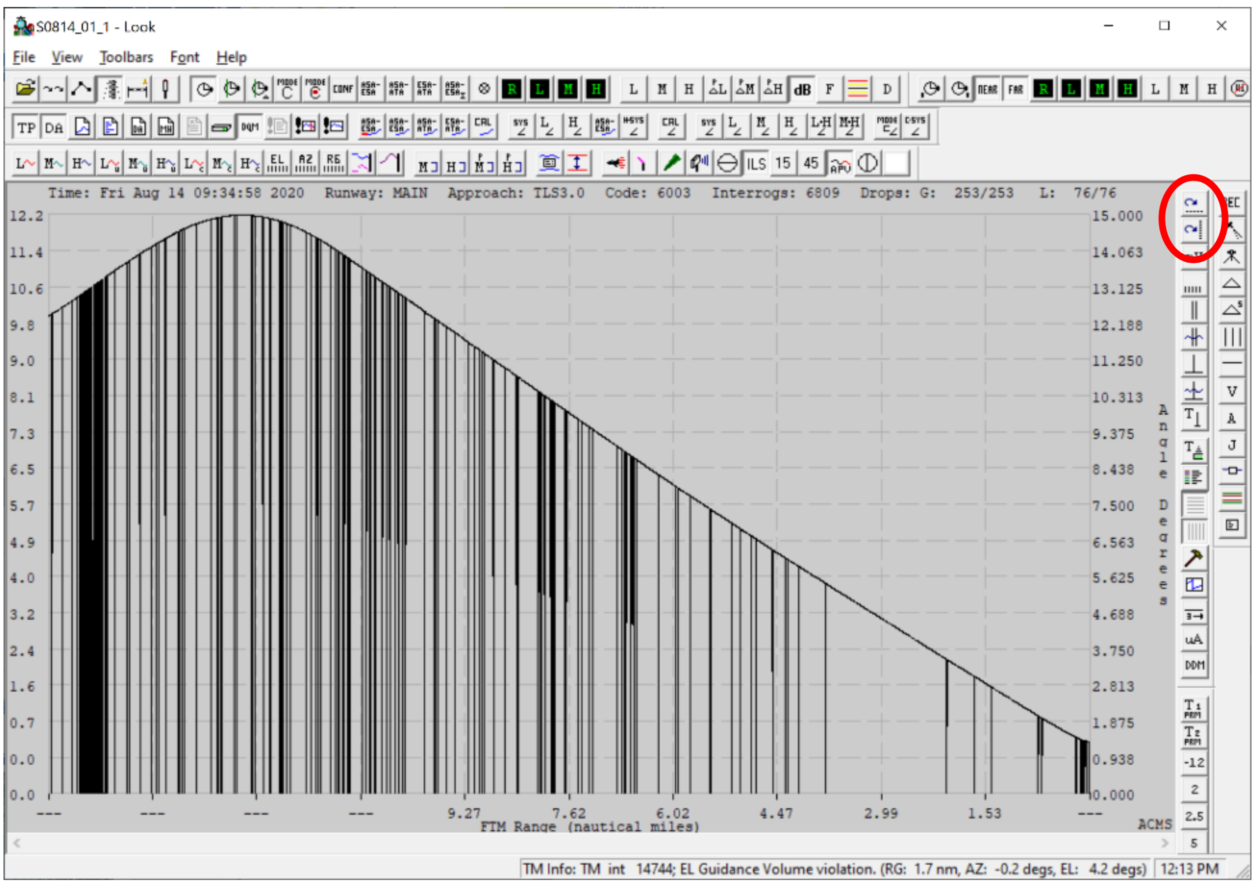

Figure 15 - Time-of-arrival (TOA) measurement during and approach

Some missing TOA measurements are shown below that are synchronous and missing for both sensors (red) and asynchronous (yellow) missing for just one sensor. The synchronous missing measurements indicate that the transponder was occupied replying to another radar or aircraft TCAS. The asynchronous missing measurements indicate possible overlap with another aircraft transponder reply or intermittent loss of line-of-sight between transponder and TLS sensor antennas.

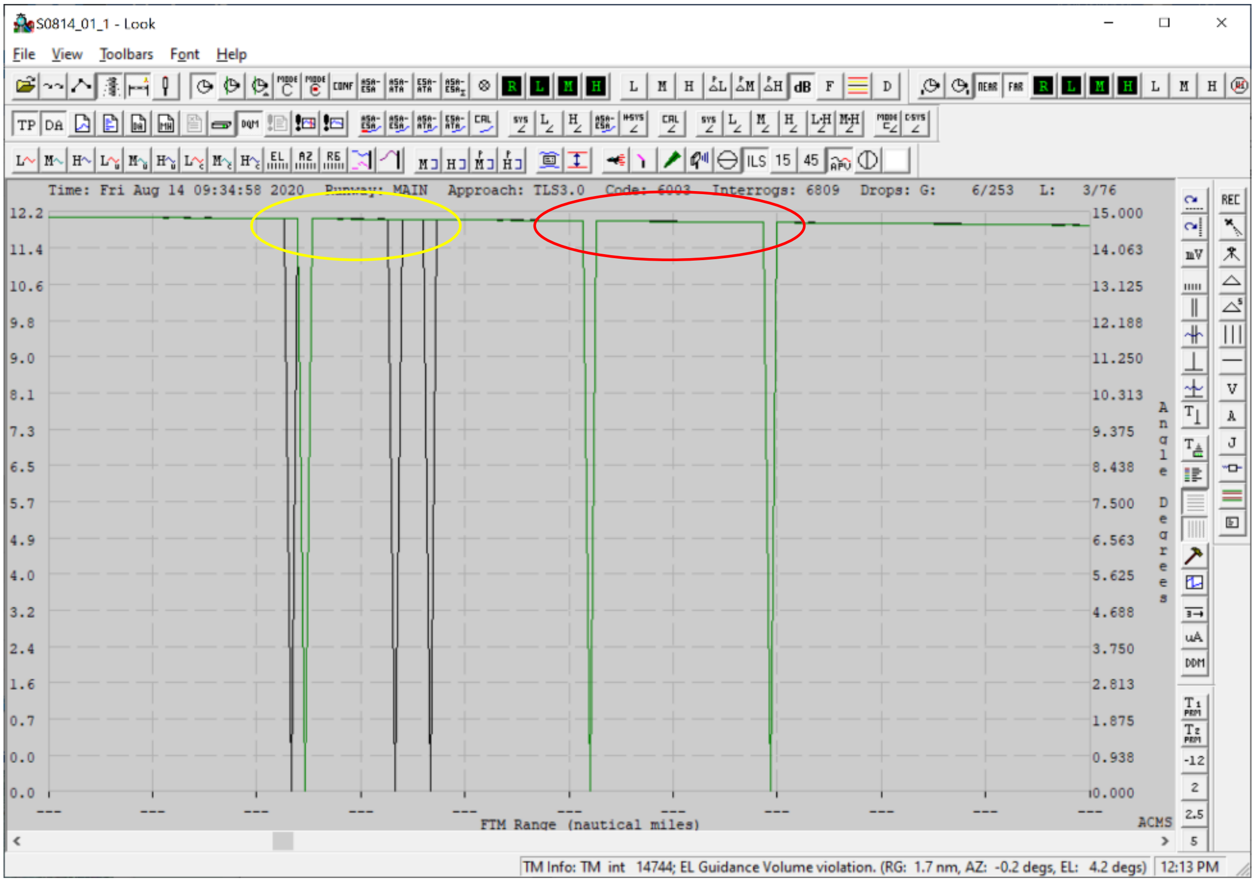

Figure 16 - synchronous and asynchronous missing measurements

In the example below, a different cause of missed TOA measurements is displayed for a fighter jet operating in a mode where the transponder is suppressed from replying to avoid interference with other aircraft systems. This reduce number of available measurements results in reduced effectiveness of the transponder tracking and is part of the reason that TLS interrogations operate at a high rate of 22 hz in order to guarantee a minimum update rate of 5 hz for the precision approach function.

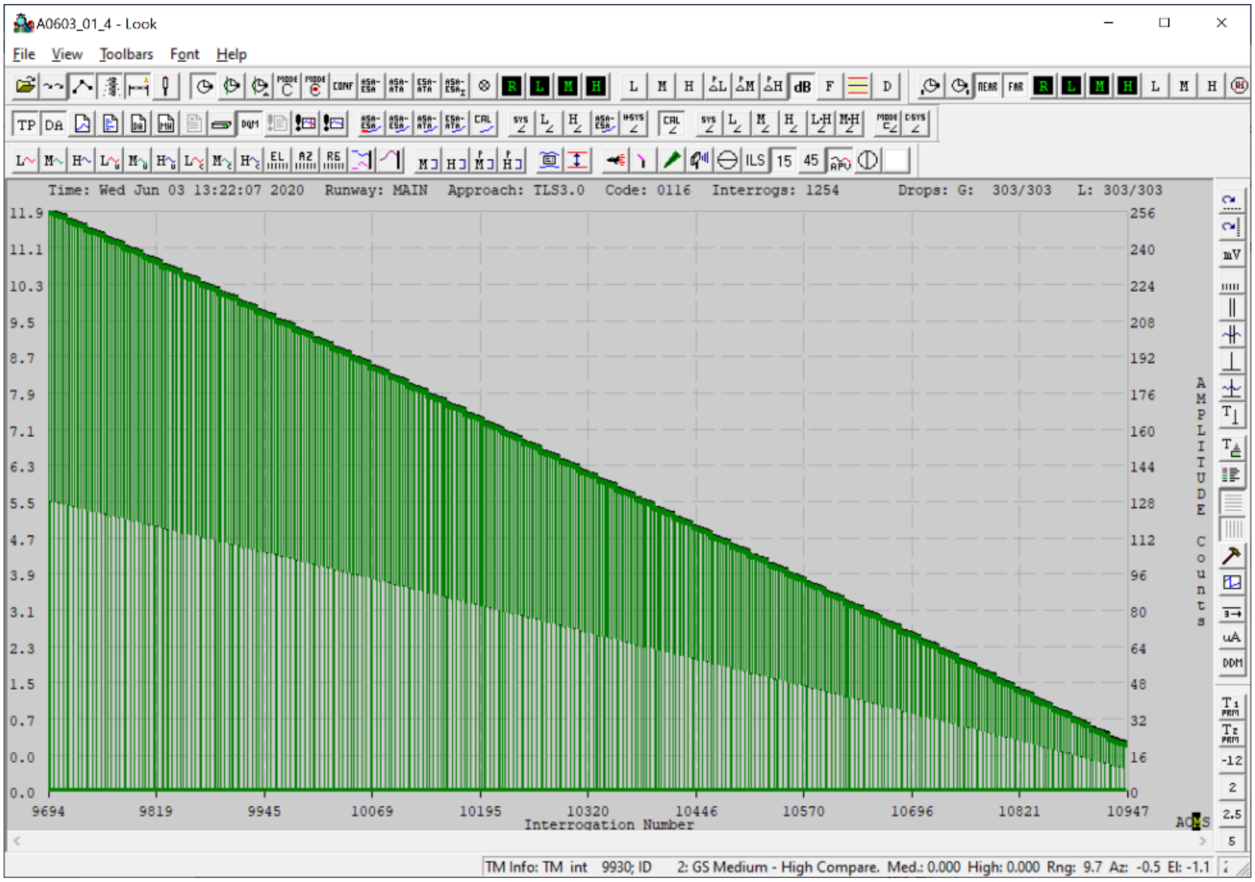

Figure 17 - Some military aircraft transponder modes reduce effective update rate

After zooming the display on the look tool (left click down, drag right, left click up) it is possible to observe that all of the missing TOA measurements for the fighter jet example are synchronous, missing from both the ESA and ASA and is at a periodic repeating rate. This indicates that the cause is within the aircraft or its transponder. Press double left click to zoom out.

The TOA measurement is the most basic fundamental measurement performed by TLS, if the TOA is missing then all the other measurements will also be missing including the amplitude, phase and frequency.

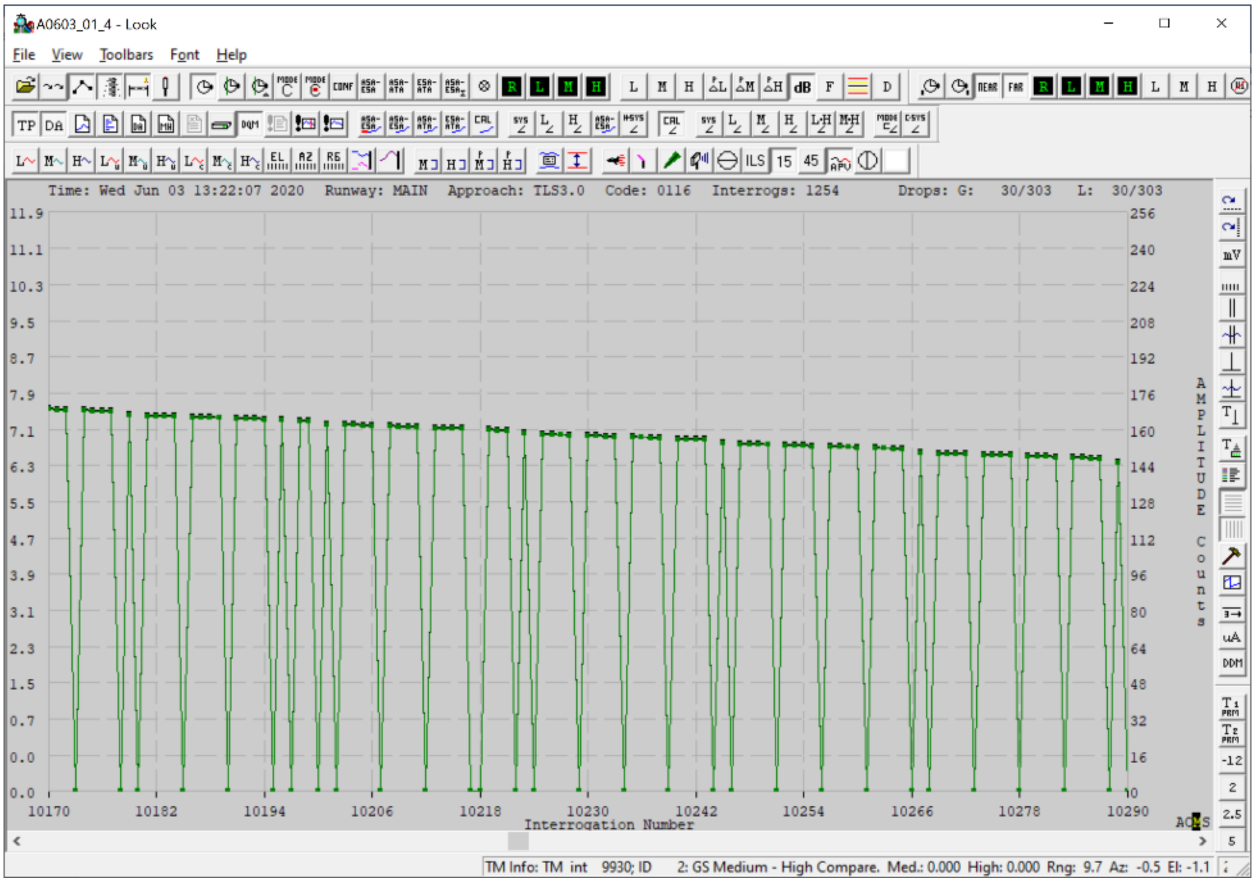

Figure 18 - reduce transponder reply rate for a military aircraft (in this case a Mirage at Assab)

Phase Measurements

TLS uses differential carrier phase tracking, or normally for TLS referred to as just “phase measurements”. The phase angle between two antennas can be displayed using the Look tool. The figure below shows the low-resolution phase measurement (brown) during an approach and the sign convention corresponds to a side view of the approach where the right vertical axis is angle above the horizon from the ESA antenna. The default units for the vertical axis are units of counts based on the 8bit digital measurement which provides a range from 0 to 256.

The first important observation from this plot is that the flight path of the aircraft can immediately be recognize, since this low-res phase angle is essentially a side elevation view of the aircraft, the aircraft rises in angle (red ellipse) as measured from the TLS ESA. The areas highlighted yellow show the repeating cycle characteristic inherent to interferometry and phase measurements.

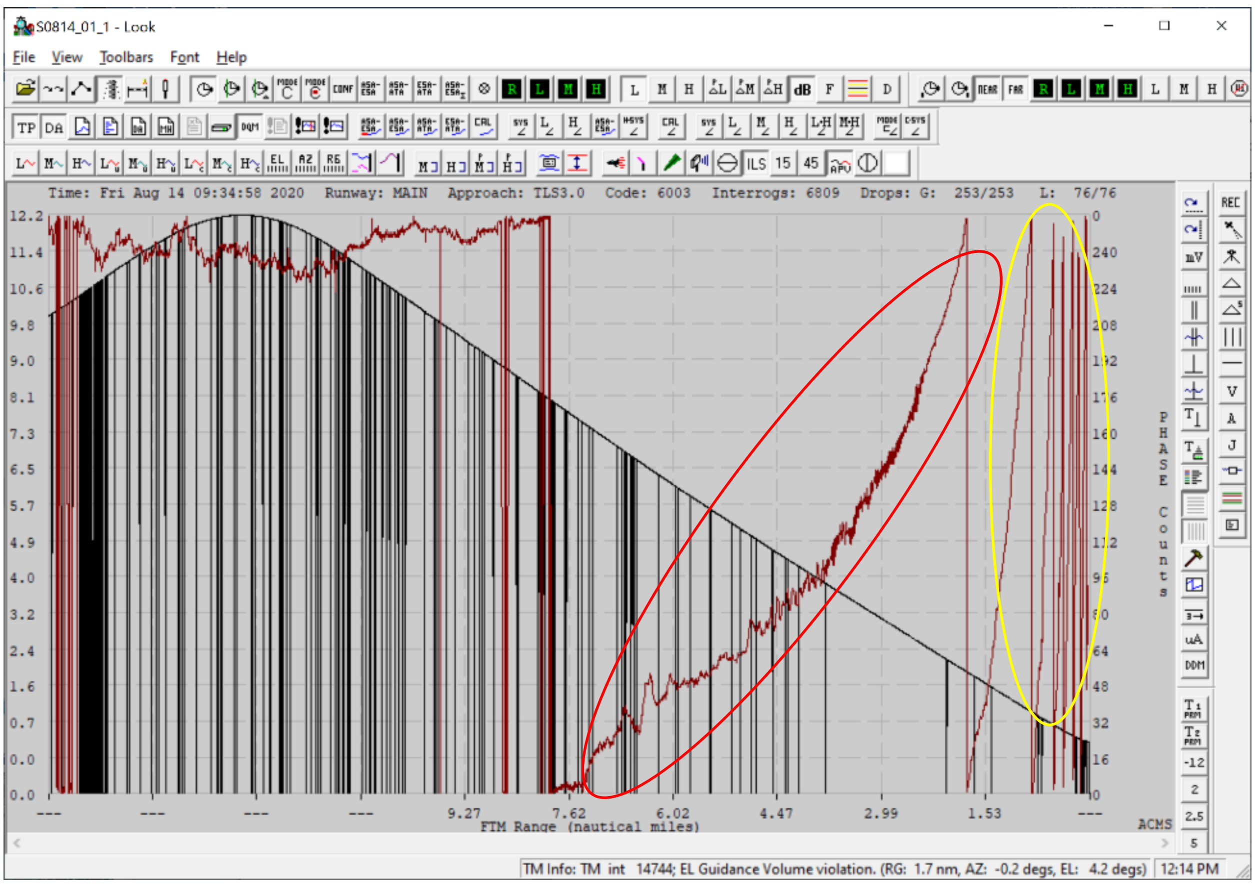

Figure 19 - example of ESA Low-res phase measurement (red ellipse)

The plot below shows the low-res track (red circle) that results from the TLS SW processing the low-res measurements (brown) and solving for the cycle (red). Note that the right vertical axis default units are now in degrees.

At 6 nm from threshold the aircraft is at about 2.8 degrees elevation measured from the ESA low resolution track (brown line), these points are highlighted with the blue circles.

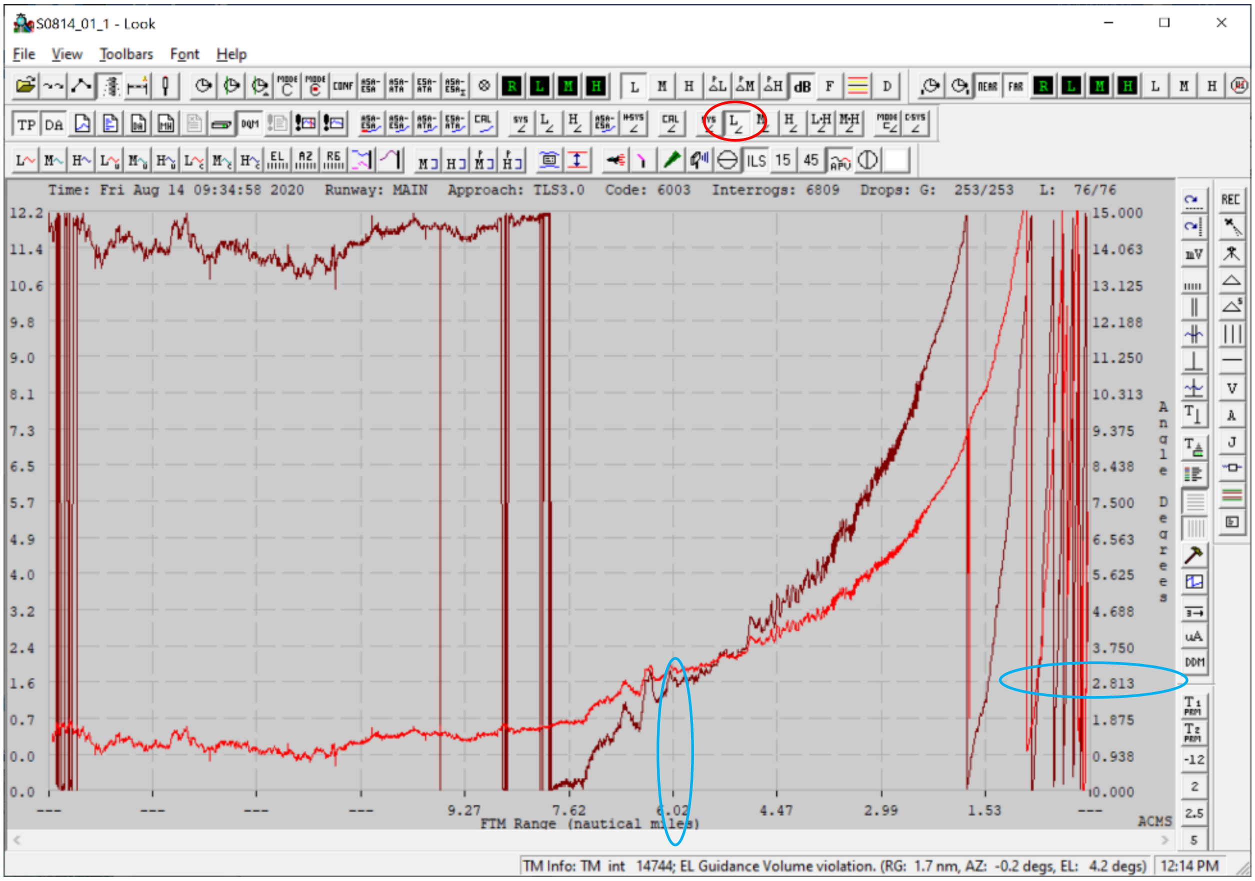

Figure 20 - Low-resolution ESA track

The high-resolution measurement displayed below can be selected with the H button (red), this is normal data for a level crossing conducted for calibration.

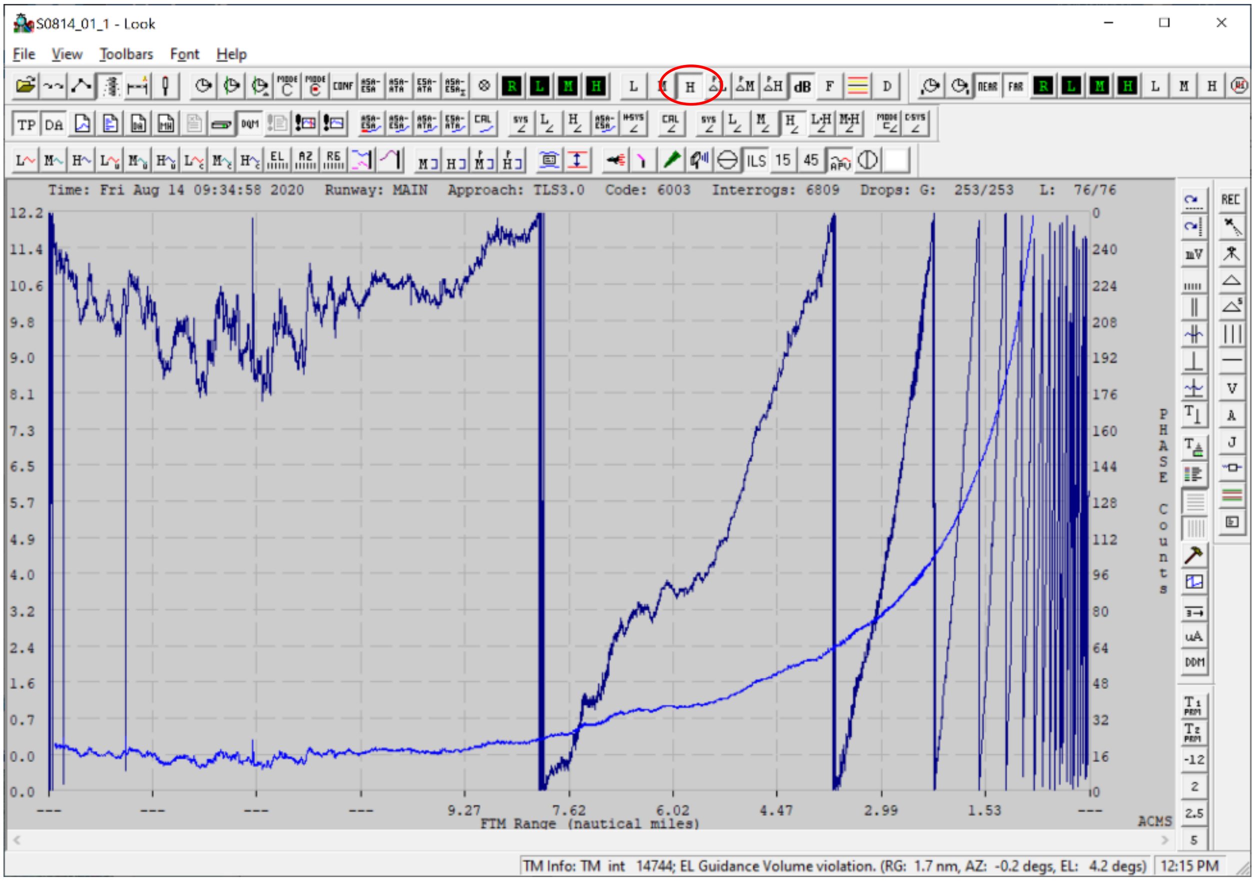

Figure 21 - high resolution raw phase measurement and the resulting track

TLS SW algorithms process the ESA low, med and high-resolution measurements and the resulting track is referred to as the system track can be displayed using the "sys" button (red). The sys track shows the aircraft starting low in elevation angle and then gradually rising in angle as the aircraft flies toward the TLS. The aircraft is actually flying at constant elevation, but the elevation angle is increasing as measured from the TLS ESA. Notice that the cycles are no longer displayed as the tracking algorithm has resolved all cycle ambiguity and has determined the vertical angle location of the aircraft.

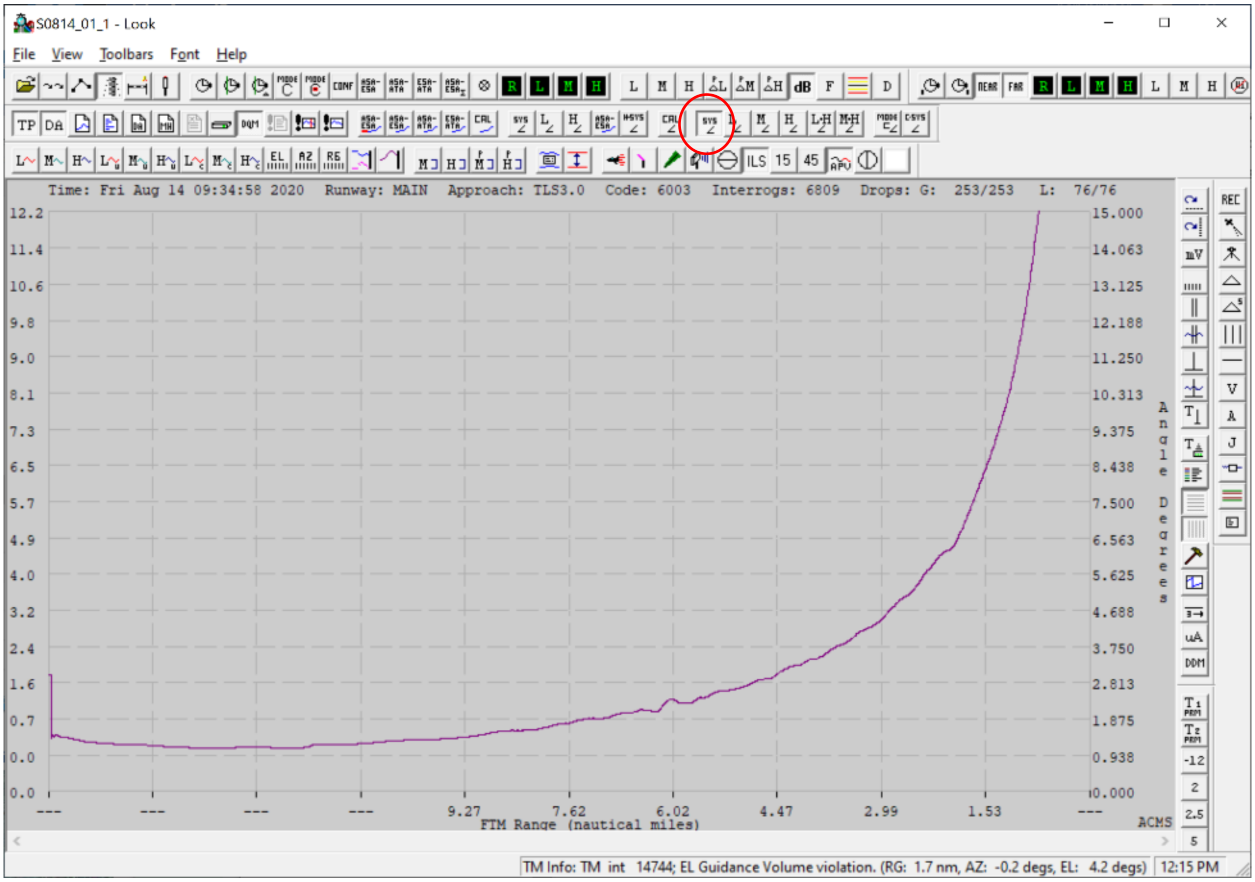

Figure 22 - The system Track that results from the ESA phase measurements

The best way to calibrate TLS is by using a theodolite as described in the TLS Calibration manual 020-0071. The theodolite data is available that corresponds to the raw file shown below, theodolite data is stored in a .dif file by the FTM SW tool. The theodolite measurement (red) was used to calibrate the TLS and this plot shows the result that can normally be expected: the TLS sys track for elevation (magenta) should overlay the theodolite (red) after the calibration process is complete. The area circled blue shows where the theodolite operator encountered the upper stop on the vertical angle adjustment of the theodolite, this is normal and does not negatively affect the calibration process.

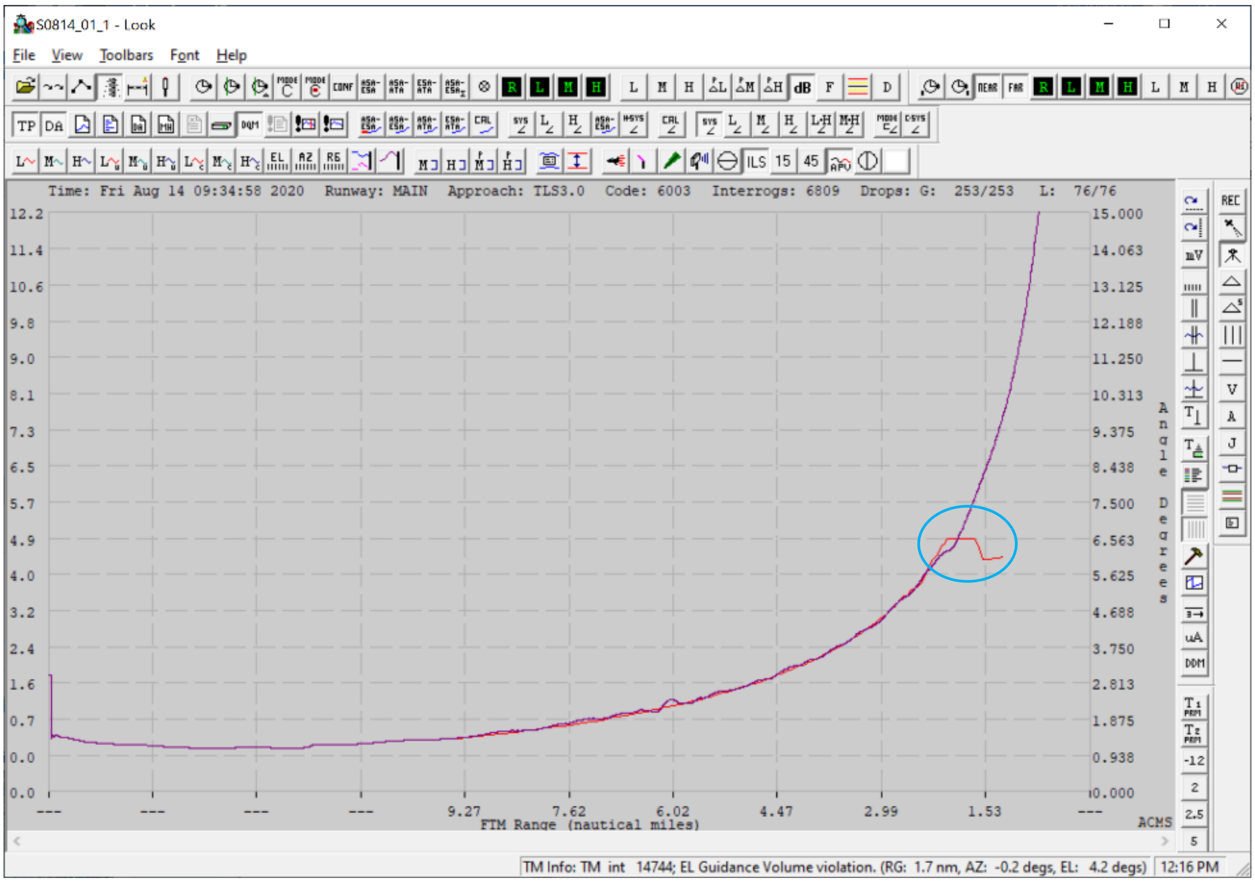

Figure 23 - Theodolite elevation measurement

A useful way that the Look tool can be used to verify the success of the calibration is to display the ESA measurement tracks with the “cal” button disabled (blue), and then select the “cal” button to observe that the measurement tracks are as close as possible to the theodolite trace.

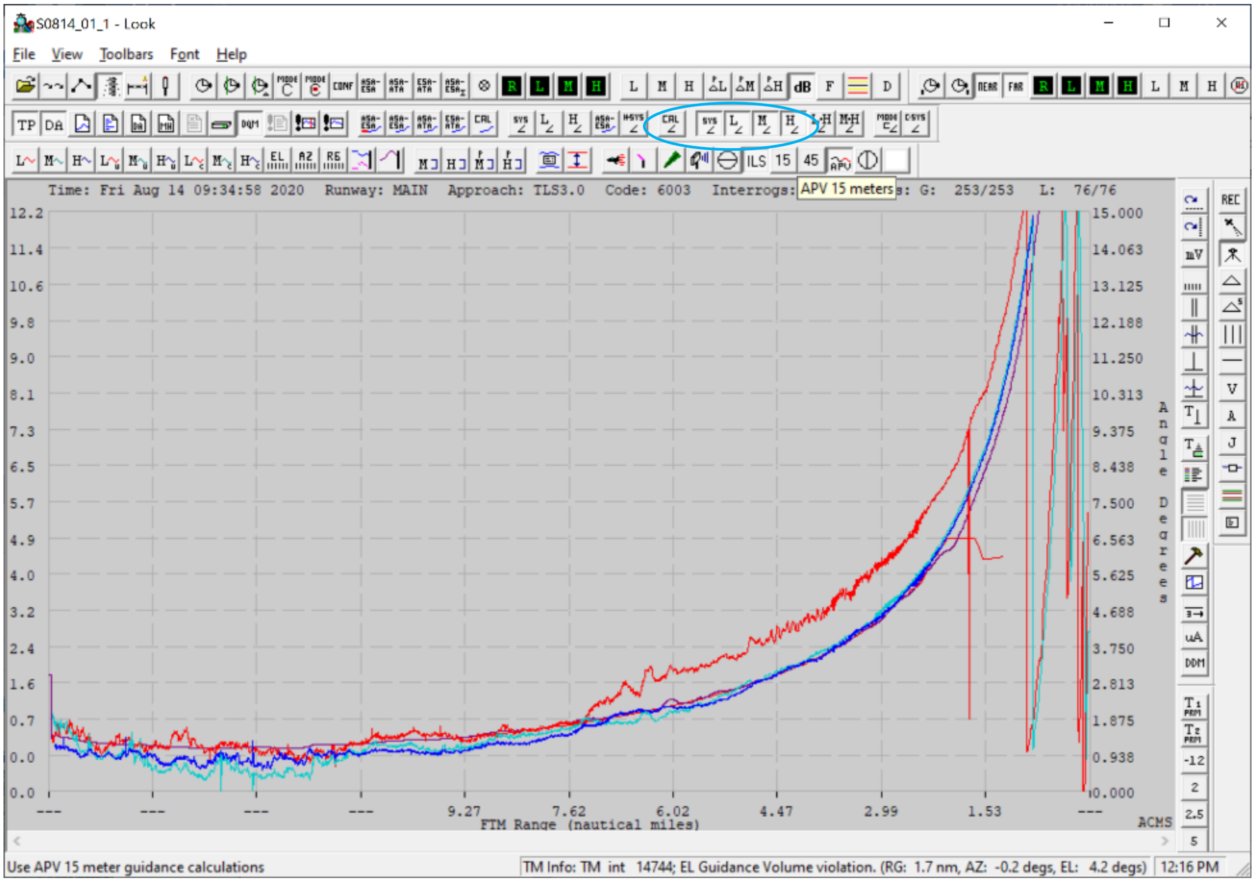

Figure 24 - example calibration data verification – cal offset not applied

With the “cal” button (blue) selected, the calibration table stored in cal.dat is applied. All of the ESA track measurements overlay the theodolite and this plot is representative of the best result that can be achieved.

Note that there are erroneous cycle slips above the upper edge of the guidance service volume of 13 degrees. Data above 7 degrees is outside the scope of the standard precision approach requirements and does not normally need to be resolved. In the case of higher glide slope angles, the user will need to make adjustments to achieve the required performance in those cases.

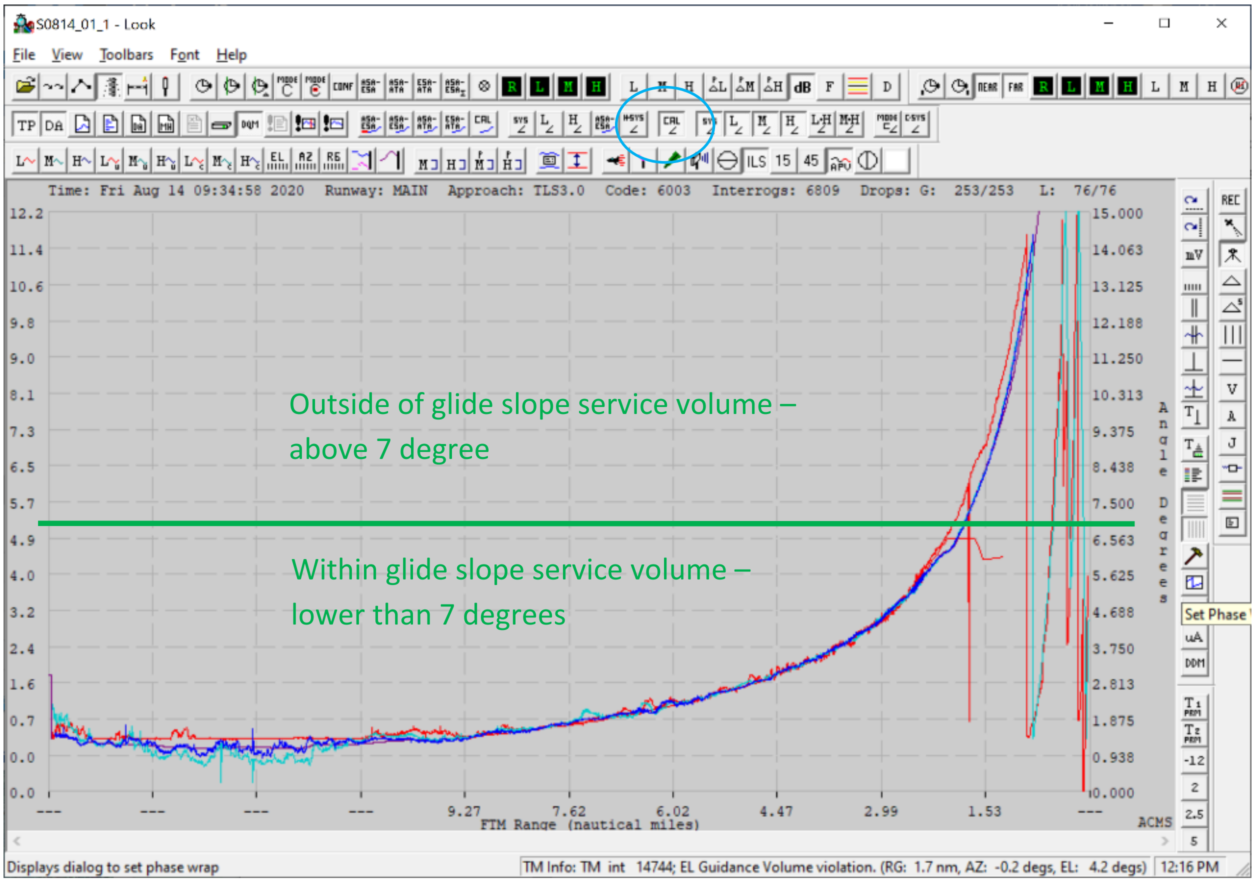

Figure 25 - example calibration data verification – cal offset applied

Mode-C

The Mode-C reply from the aircraft transponder can be displayed with the mode C button (circled blue). Mode-C can provide a second independent verification of the calibration that was completed with the theodolite. Also as shown in the TLS Calibration manual 020-00071, Mode-C can be used to calibrate TLS. The Mode-C measurement from the aircraft is shown in magenta in the figure below.

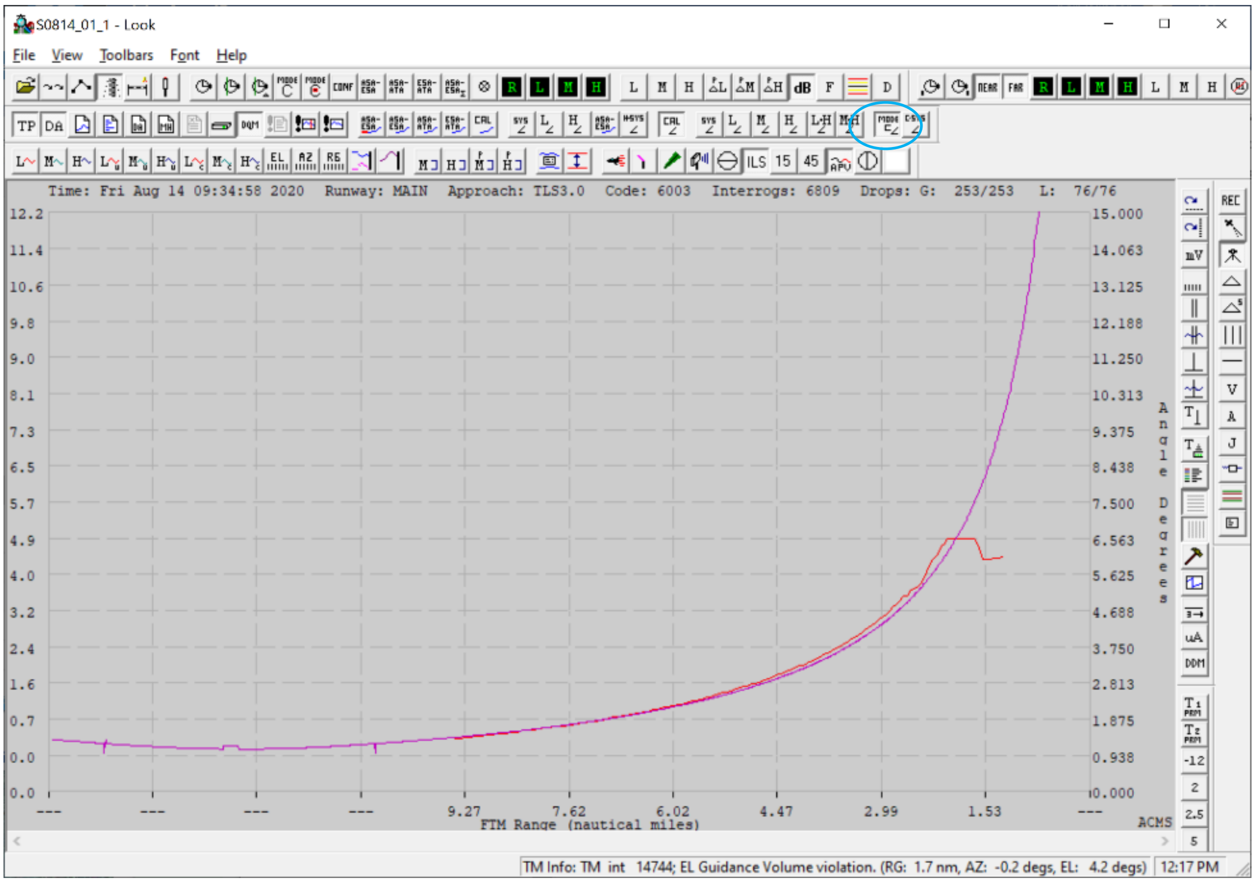

Figure 26 - theodolite data

The difference between TLS elevation track and the Mode-C is very useful if there is a doubt about the glide slope angle, the plot below confirms that TLS glide path is set to 3.0 (0.0 on this scale is the centerline of the glide path which is 3.0 degrees above the horizon.)

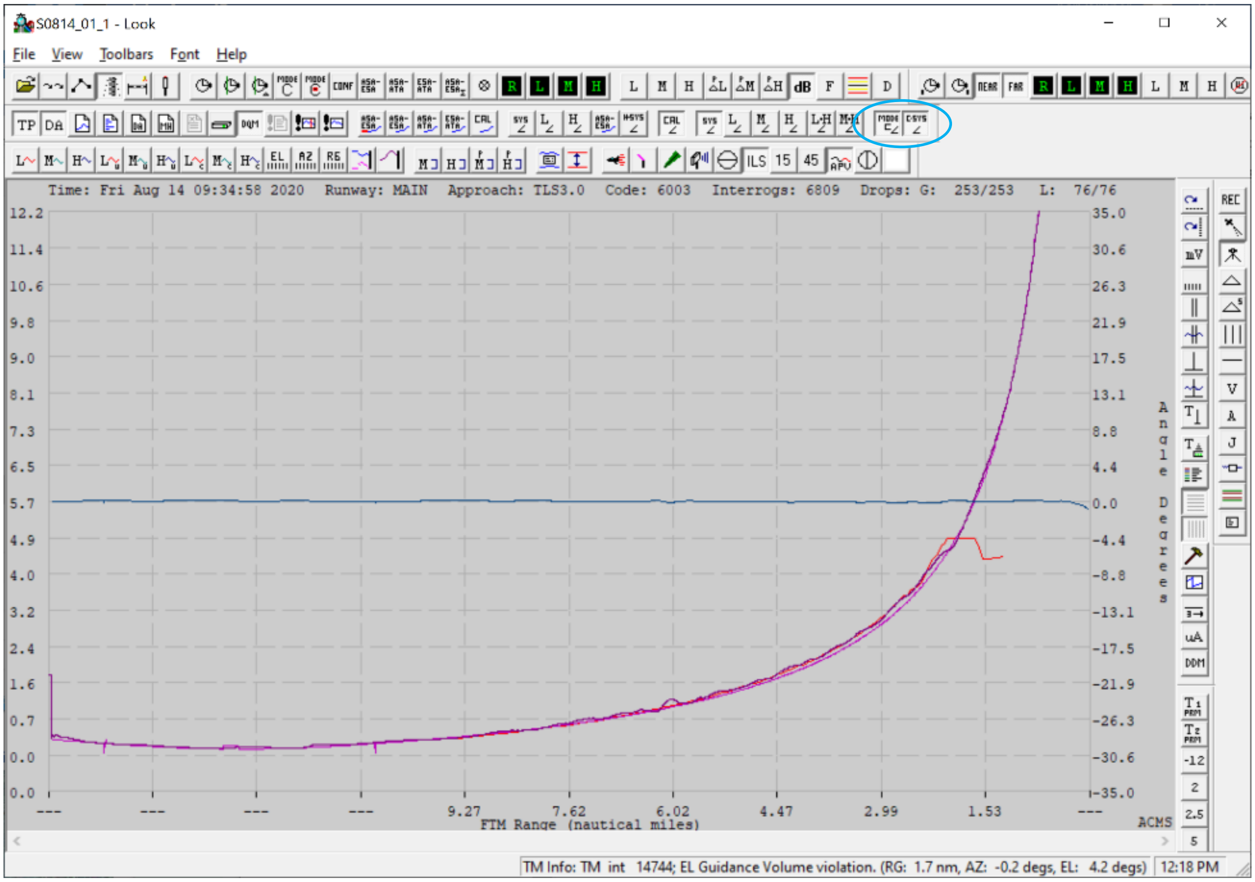

Figure 27 - comparing to Mode-C

GTU transmission

The glide slope needle command that was transmitted to the aircraft can be displayed. In the figure below, the commanded needle deflection for full scale fly up is transmitted from beyond 6 nm and starting at 6nm the commanded glide slope needle becomes proportional to the angular distance from the glide slope. At approximately 4.47nm from the runway threshold, the aircraft was at 3.0 degrees and then continues above the glide slope and the resulting transmitted needle command is a fly down signal.

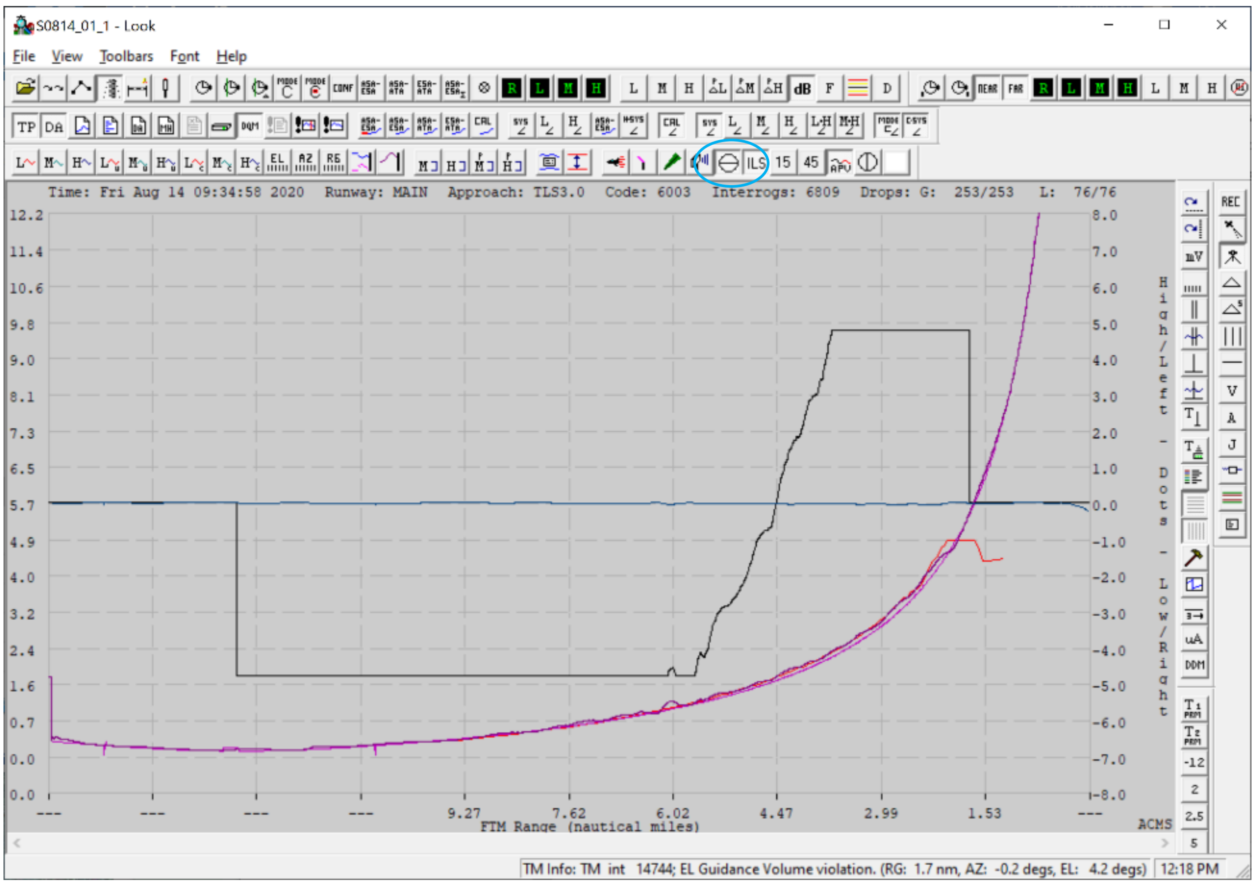

Figure 28 - checking the linearity of the glide slope needle (black) during a crossing

The glide slope needle transition from full scale fly up to full scale fly down should be as linear as possible. The linearity of this plot is a reflection of the calibration quality. Flight inspection measurements that depend on this linearity are path width and symmetry. Also, the flyability of the approach and stability of auto couple approaches will be optimal with this transition as linear as possible.

Interrogation rate

The TLS interrogation rate setting should be at least 22 hz, In the figure below, the updates rates for azimuth, elevation and range (all circled red) can be inspected to verify normal tracking subsystem update rates.

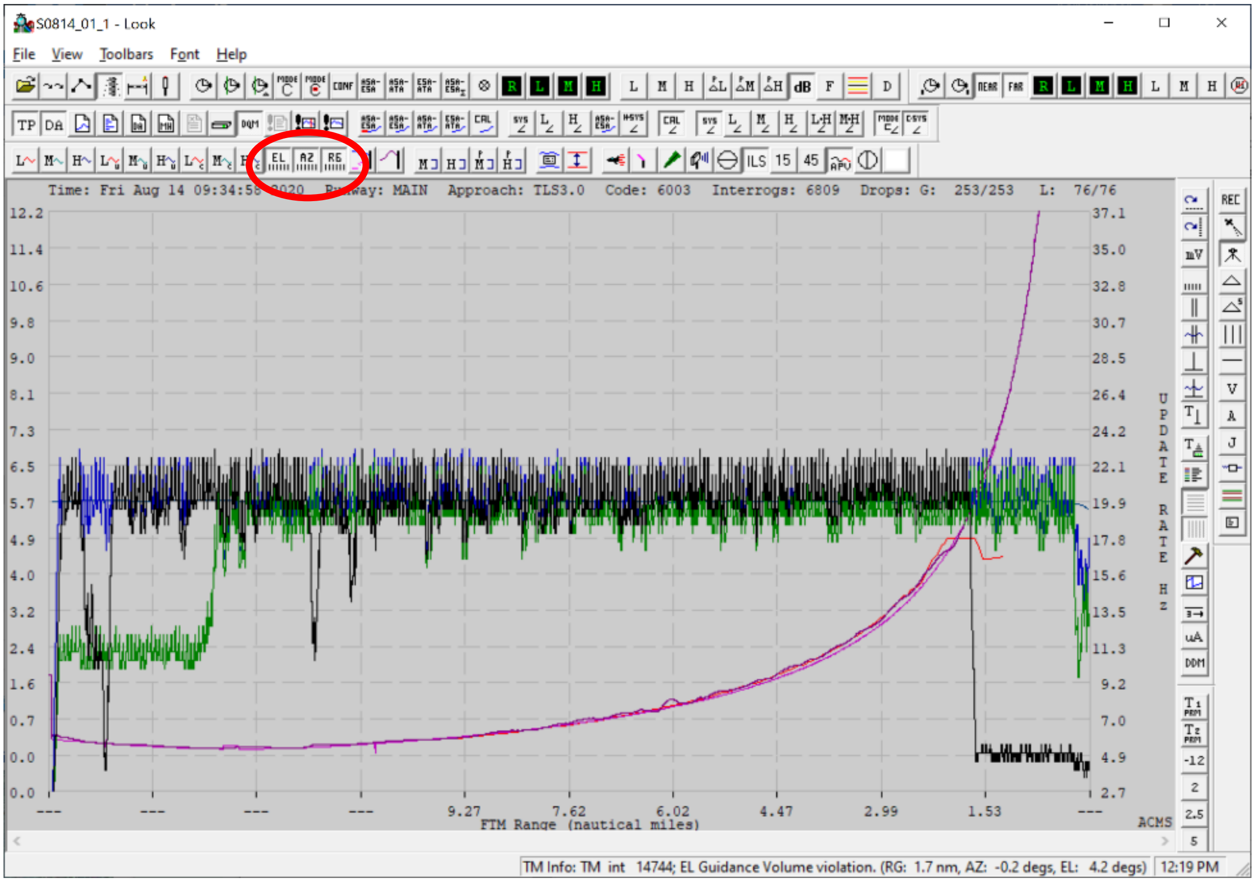

Figure 29 - Normal updates rates range from 15 to 22 hz.

Phase measurements and tracks can also be inspected for the ASA. The figure below shows typical phase track data for an approach.

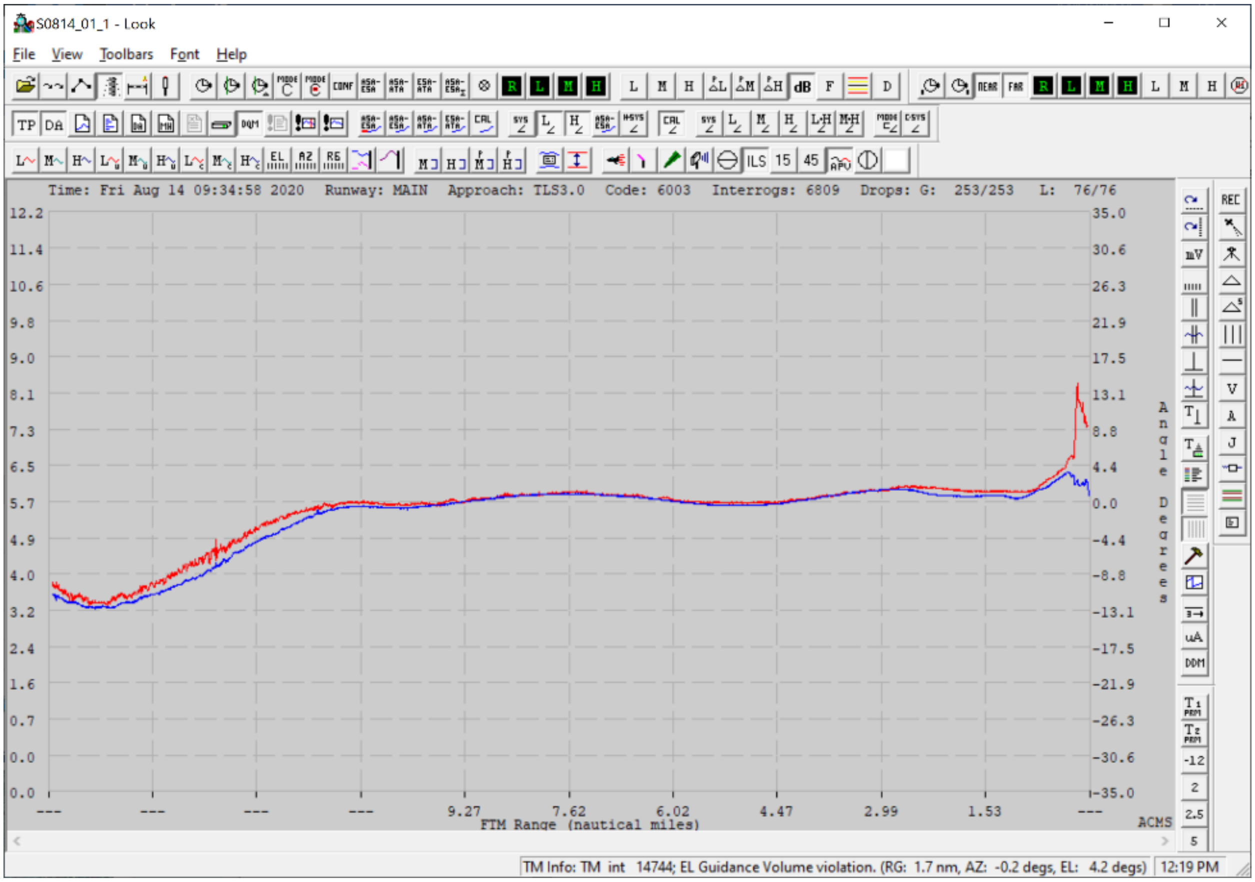

Figure 30 - typical ASA phase tracks for the L and H

The figure below shows an example of and approach when the glide slope was disabled twice during one approach (circled red). Initially the glide slope was off beyond a range from threshold of 12nm which is the normal maximum range for the glide slope. Then a fly up indication was broadcast followed at about 10nm and again at 8.5 nm by the glide slope transmitted DDM returning to zero.

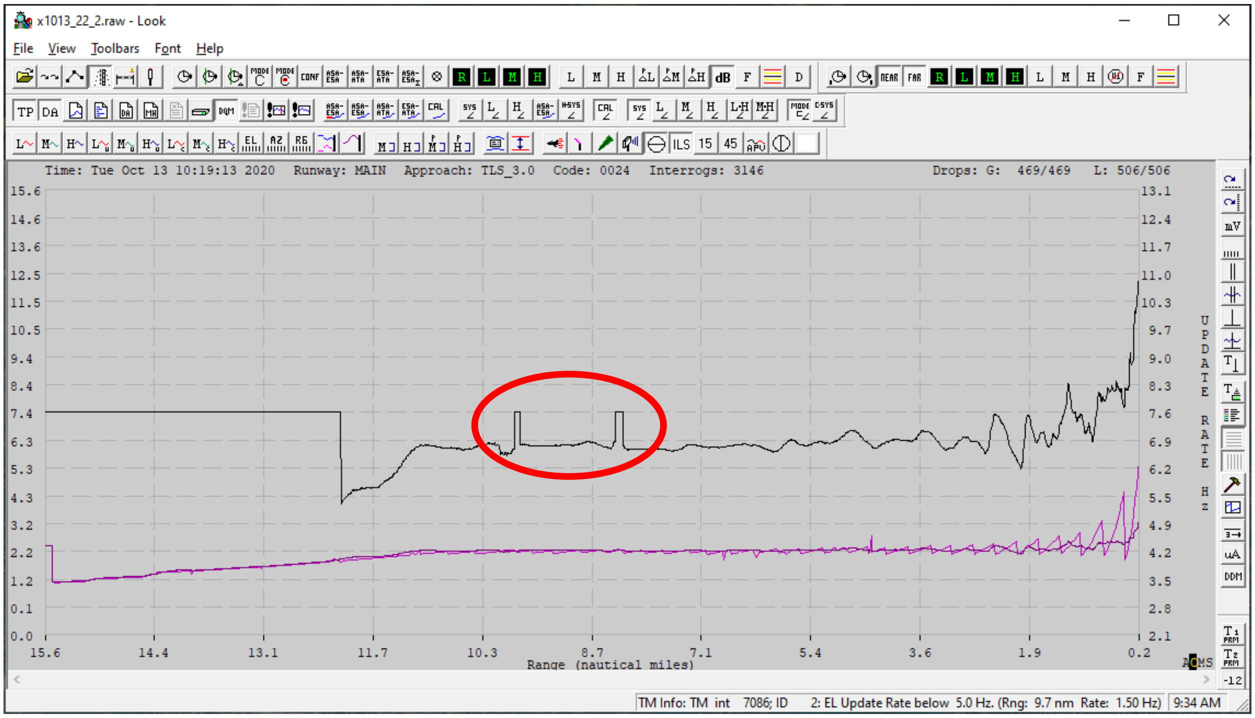

Figure 31 - loss of glide slope anomaly

By investigating the two dropped glide slope events further, we find that the update rate dropped to near zero during the two events above.

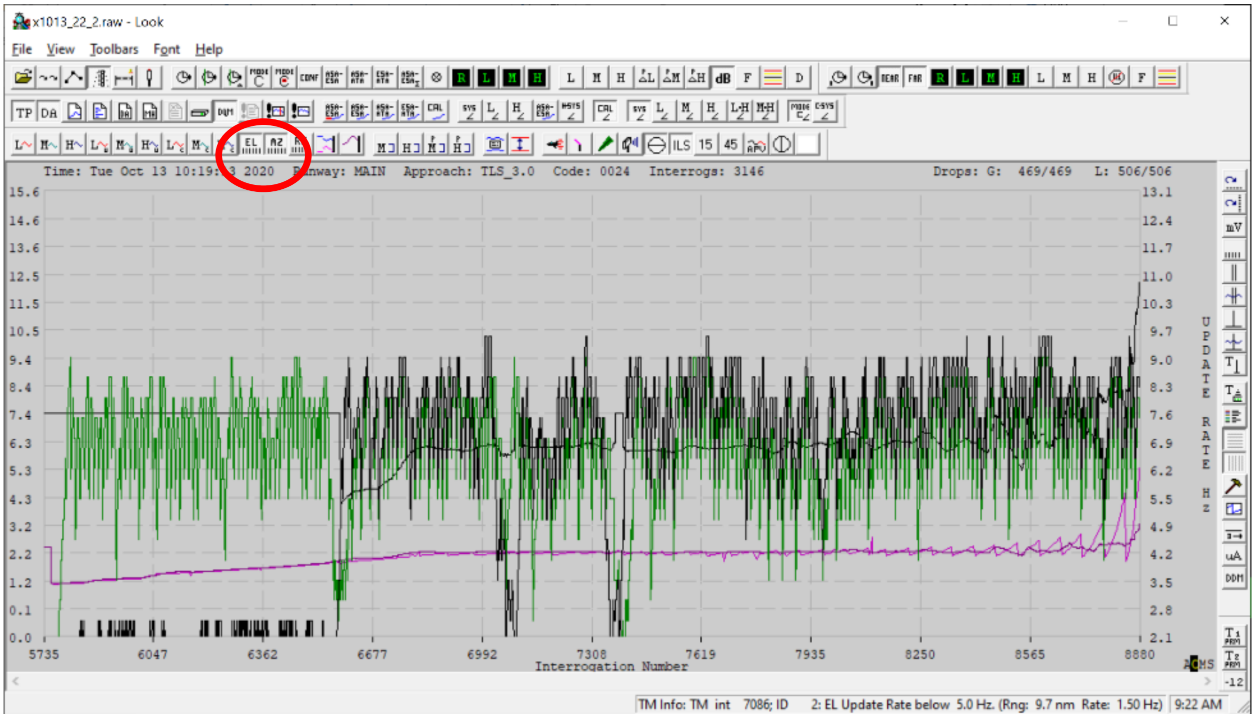

Figure 32 - poor update rate caused by absence of P2 and improper update rate setting

Update rate was averaging about 7 hz and reduced to near zero hz on two occasions that correlate with the glide slope signal turning off.

The corrective action for the above situation is to increase the update rate from 11 hz to 22 hz and enable the p2 suppression pulse on the interrogation.

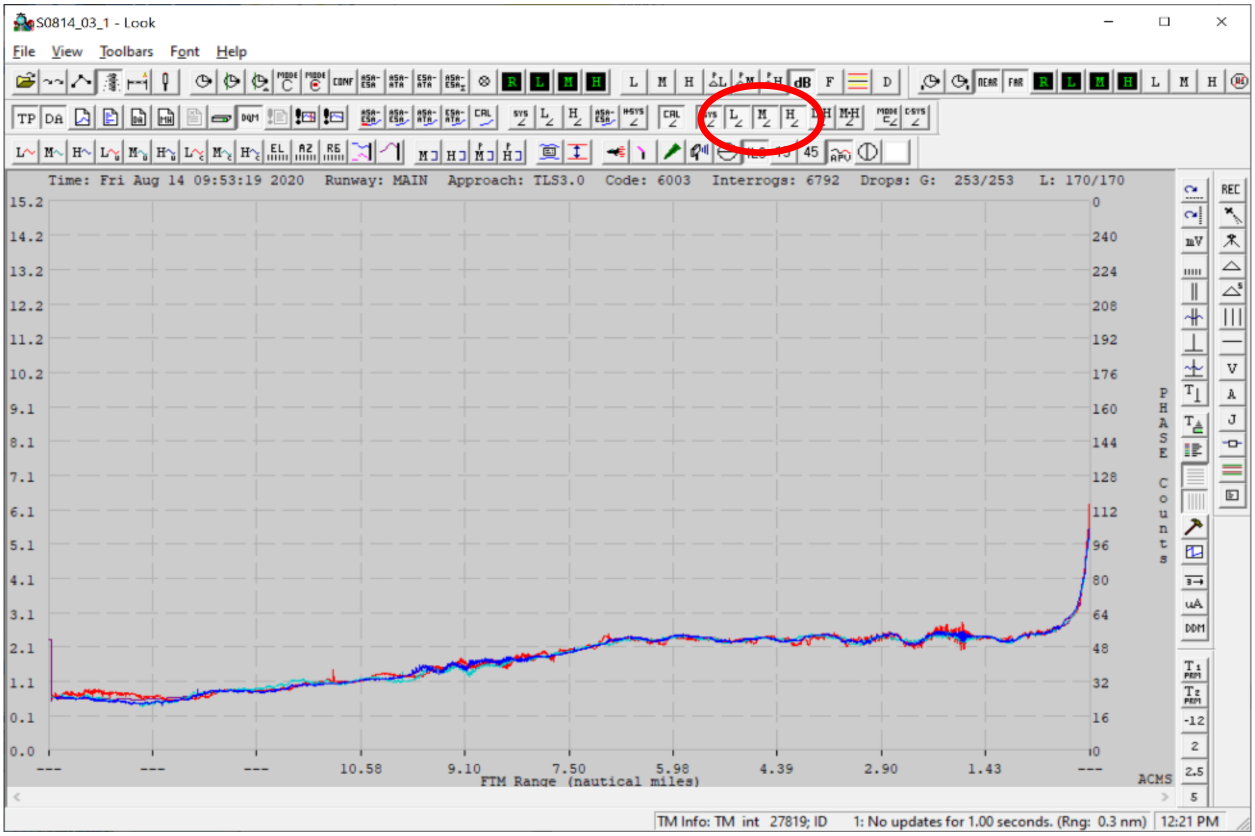

Figure 33 - normal elevation tracks for the ESA L, M and H.

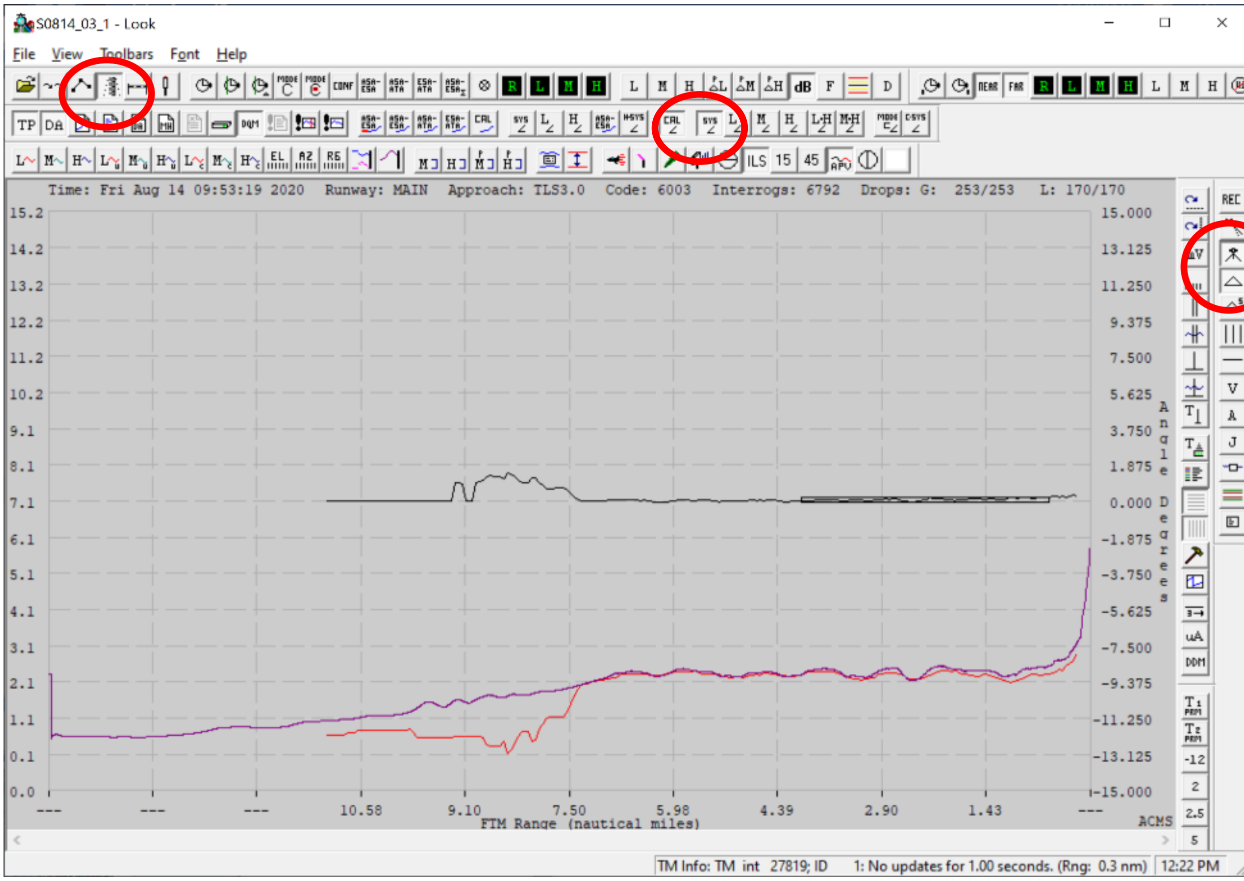

Figure 34 - Using Look, it is possible to display the difference between the Theodolite and the system elevation track for an approach compared to the ICAO tolerance

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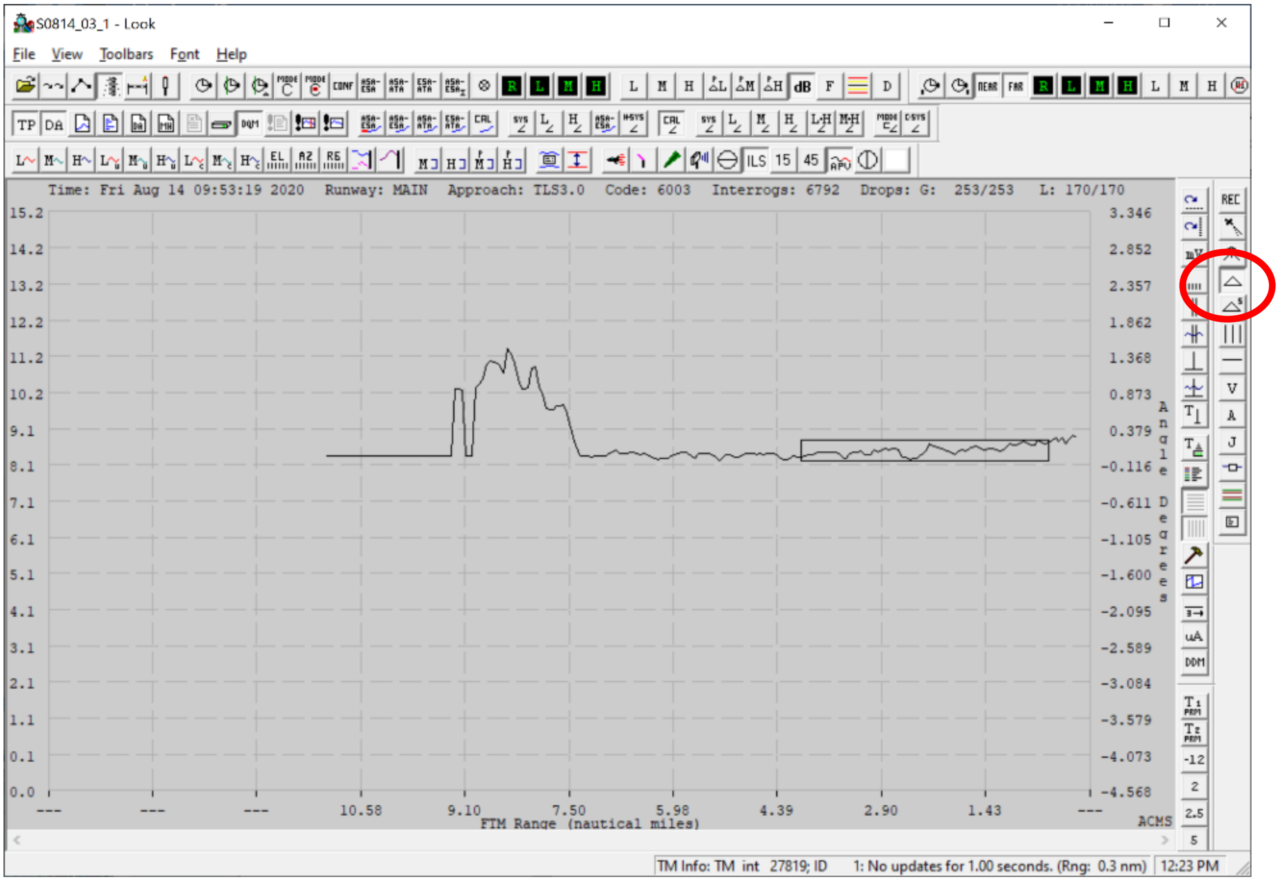

Figure 35 - theodolite evaluation of glide slope track error

Using Look mouse clicks to zoom in on the theodolite trace, it is possible to display the difference between the Theodolite and the system elevation track for an approach compared to the ICAO tolerance.

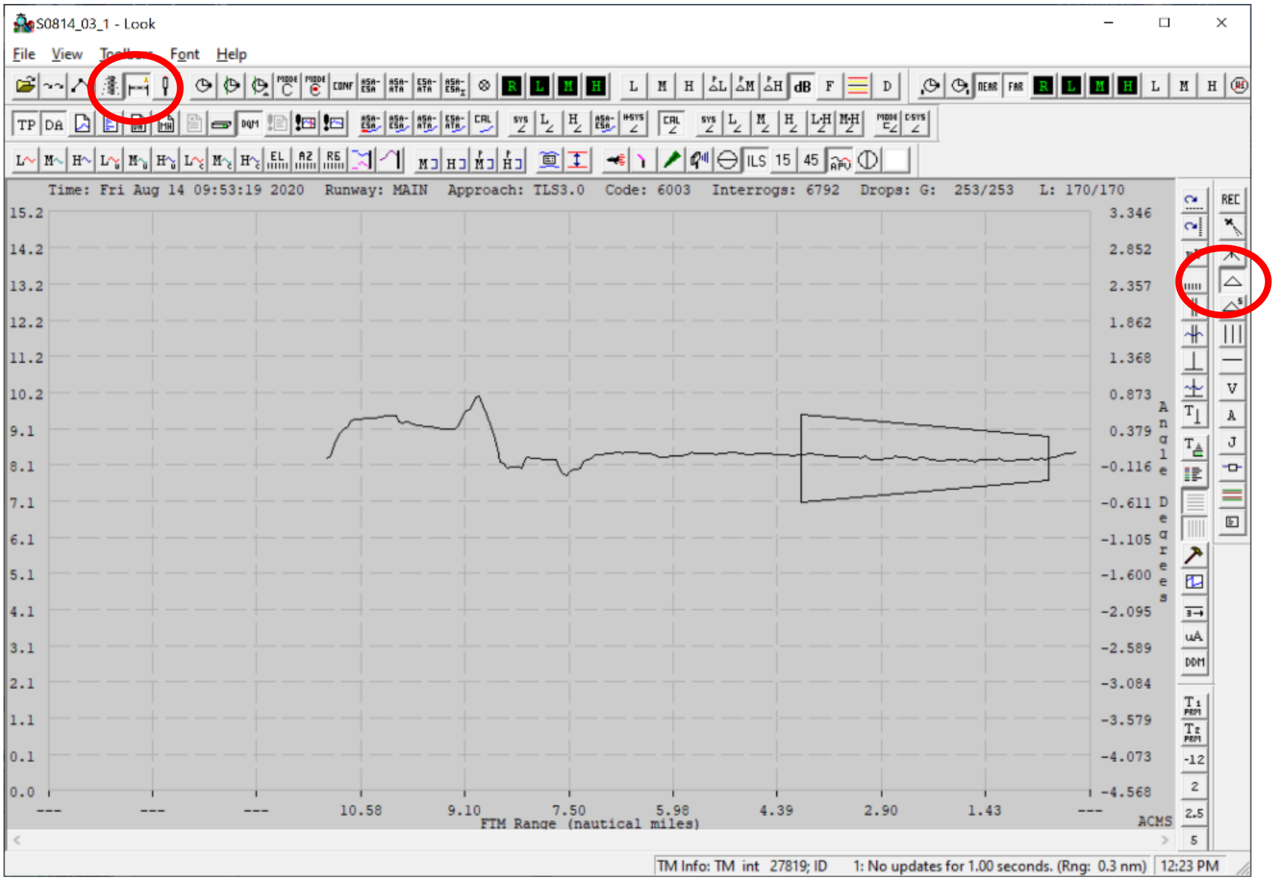

Figure 36 - theodolite evaluation of azimuth track error

Using Look, it is possible to display the difference between the Theodolite and the system azimuth track for an approach compared to the ICAO tolerance.

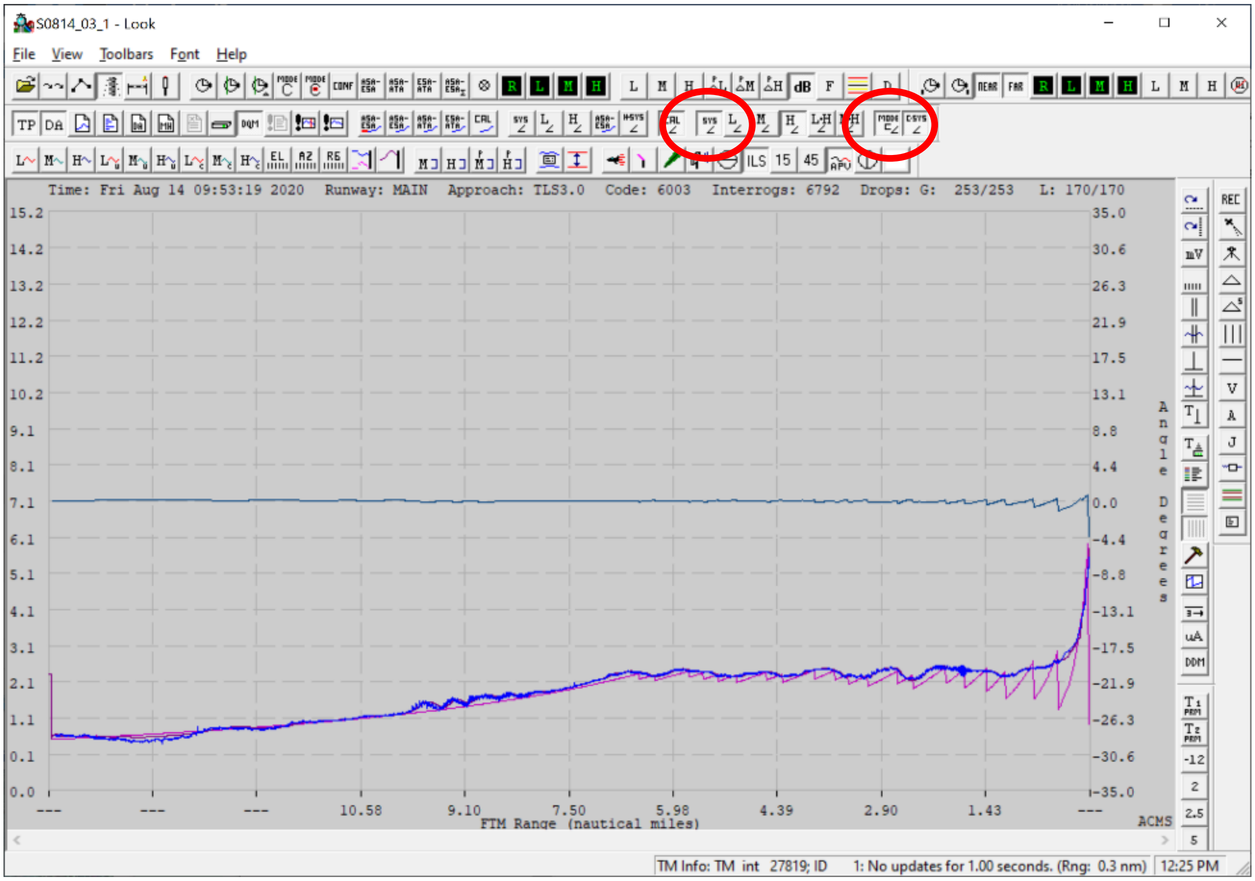

Figure 37 - If theodolite data is not available for an approach, the Mode-C response from the aircraft can be displayed to verify the glide slope angle.

The aircraft's Mode-C equipment has a resolution of 150 feet and so comparing the Mode-C elevation to TLS at the furthest practical range is best. In the figure above, at 7.5nm for example, 150 feet / 7(6080)feet yields a mode-C angular resolution of 0.2 degrees. For military use of TTLS, this alignment accuracy has been accepted as sufficient by the customer (at Assab and Al Dhafra, UAE)

Using the Look tool, it is also possible to inspect BIT data during an approach. In the figure below, the BIT for M and H phase can be inspected during the course of an approach.

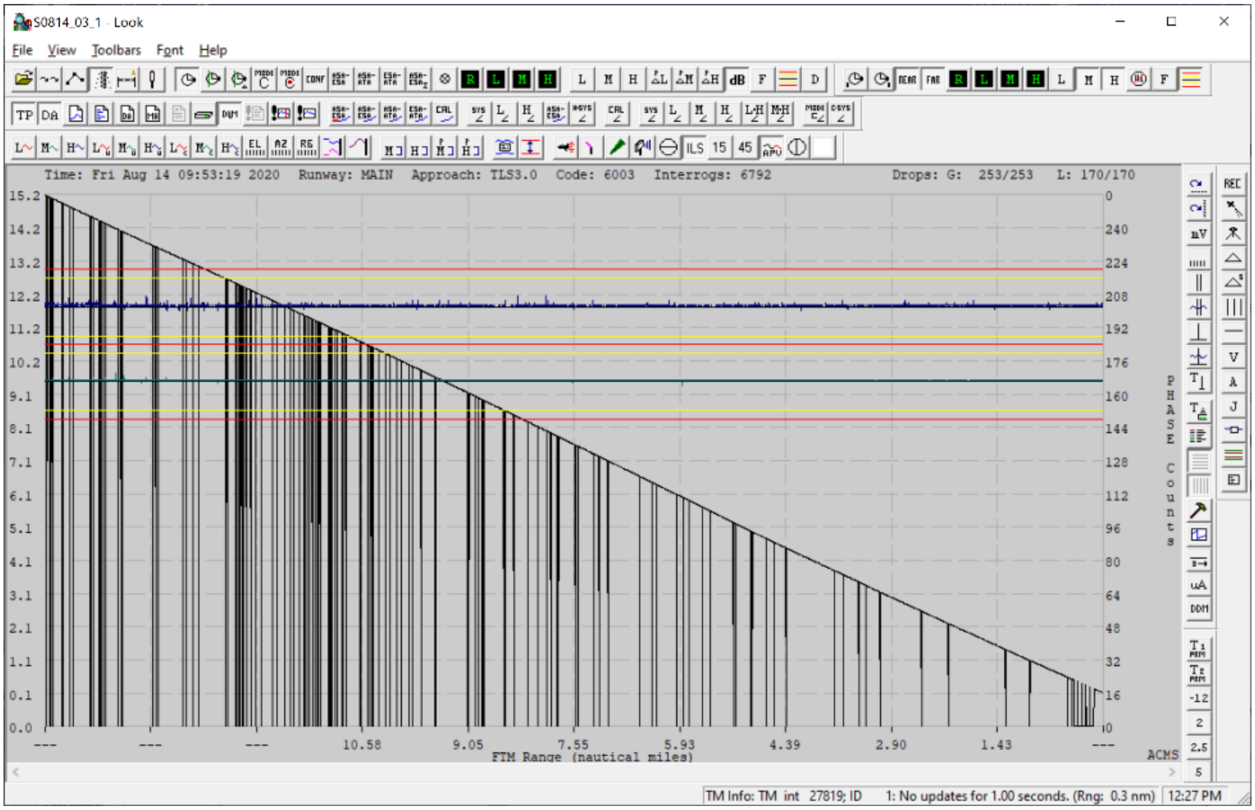

Figure 38 - ESA M and H phase during an approach.